Real Analysis I

Proofs, Structures, and Counterexamples in Calculus

Self-contained lecture notes for a 14-week course

Cumulative release — through Unit 6

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These notes use Stephen Abbott's *Understanding Analysis*, Chapters 1–5 and 7, as the basis for their core sequence of topics. The exposition is organized around proof construction: making quantifiers visible, choosing witnesses, tracking hypotheses, and learning when a theorem cannot apply. Counterexamples are treated as part of the core theory rather than as curiosities; they identify a failed claim, the hypothesis or conclusion responsible for the failure, and a corrected statement.

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How to Use These Notes

These notes are self-contained at the level of the course. Abbott remains the primary textbook source for its motivating investigations, alternate arguments, and extensive exercise collection. The present manuscript supplies a lecture-ready sequence of definitions, theorems, and proof strategies, together with graded proof practice and systematic counterexample analysis.

About this cumulative release. This student edition contains Units 1 through 6 in full. Later units are withheld until their release week. The Course Map remains visible as a semester roadmap; unreleased rows are muted in gray. The appendices and back matter are included in every release. References to later theorems in the unit spine are rendered descriptively until the target unit is released, so an early edition never contains broken forward references.

The course repeatedly asks four questions.

1. **What does the definition literally say?** Write the quantifiers, their order, and the permitted dependencies before manipulating symbols.
2. **What is the proof strategy?** Decide whether the argument is direct, by contraposition, by contradiction, by induction, or by construction.
3. **Where is each hypothesis used?** Apply a theorem only after verifying each of its hypotheses, and record where those hypotheses enter the argument.
4. **What fails if a hypothesis is removed?** Build or recall a counterexample that isolates the missing assumption.

Course shorthand: the epsilon principle. Throughout these notes, this phrase denotes the following recurring argument: if a nonnegative quantity is smaller than every positive tolerance, then it must be zero. To prove this, assume the quantity is positive and choose an epsilon smaller than it. The same argument will reappear in uniqueness proofs, supremum approximations, limit proofs, continuity estimates, and upper–lower sum arguments.

The intended learning cycle is

$$\begin{aligned}\text{example or puzzle} &\longrightarrow \text{definition with visible quantifiers}\\ &\longrightarrow \text{guided proof with an embedded reconstruction pause}\\ &\longrightarrow \text{counterexample and corrected statement} \longrightarrow \text{independent reuse.}\end{aligned}$$

A stable weekly rhythm

Every week follows the same broad path: a guiding question, a definition or central distinction, one proof strategy developed carefully, a counterexample reflection, two tiers of exercises, and a concise closing panel. Major synthesis weeks retain a longer checkpoint. The repetition is intentional. It lets attention stay on the mathematics rather than on navigating a new format.

Using the inquiry modules

The pale *Inquiry* cards are detachable launch modules. Each appears before the first numbered section of its unit. Attempt the prediction or construction before continuing; the resolution is absorbed into the subsequent core exposition rather than displayed as a second answer block. An instructor may assign,

discuss, or skip a card without breaking the core exposition. Green *Pause and reconstruct* panels remain embedded inside selected proofs, immediately before the decisive choice or estimate. Read to the panel, attempt the missing step, and then continue with the same proof. Their purpose is to make reusable decisions visible without creating a duplicate proof.

The inquiry modules are introduced in four batches: proof language and the construction of N or δ from a prescribed epsilon (Units 1–4); strategy and representation choice (Units 5–8); theorem hypotheses and regularity investigations (Units 9–12); and partition construction and integrability analysis (Units 13–14).

How the exercises work

Formative practice develops a known proof strategy. These problems may identify the relevant definition, theorem, strategy, or intermediate estimate. The support gradually fades within the set.

Further problems provide less guidance. They do not announce whether the appropriate response is a proof, a counterexample, a comparison, or a combination of earlier ideas. Identifying the appropriate strategy is part of the problem.

The closing panel is deliberately open-ended. Most units ask for one precise statement, one reusable proof decision, and one counterexample and its corrected claim. Units 4, 7, 10, and 14 retain a longer synthesis checkpoint that also asks for a theorem and an unresolved question.

How proof support fades

The guided support is designed to recede across the semester.

- **Units 1–4:** proof forms, quantifier dependencies, and decisive constructions are named explicitly.
- **Units 5–8:** the notes identify a model family or representation, but students must complete more of the construction.
- **Units 9–11:** selected problems ask students to choose among several plausible proof strategies.
- **Units 12–14:** some problems state only the hypotheses and desired conclusion; selecting the proof strategy is part of the task.

Formative practice may still name a strategy when the purpose is fluency. Further problems increasingly suppress method labels.

Proof standard. A proof is not complete because the intended idea is visible. It must identify arbitrary objects, state dependencies of chosen constants, justify theorem hypotheses, and end by explicitly returning to the desired conclusion.

A guide to mathematical statement labels

Mathematical writing uses several labels for different kinds of statements. The labels do not form a hierarchy of logical strength: a lemma is not "less true" than a theorem, and a proposition is not proved by a weaker standard. Instead, the labels tell you the *role* a statement plays in the exposition and what kind of justification is expected.

Label	What it does	What you should expect
Axiom	States a basic assumption of the theory.	It is accepted without proof inside the present framework. Later results may depend on it. In these notes, the Least Upper Bound Property is an axiom for $\mathbb{R}$.
Definition	Fixes the meaning of a term or symbol.	A definition is not something to prove true or false. Instead, you *unpack* it when proving facts about the object it defines.
Theorem	Records a proved result of central or broad importance.	A theorem requires a proof from earlier axioms, definitions, and established results unless its proof has been explicitly deferred.
Proposition	Records a proved result that is useful but more local or less central than a theorem.	The proof standard is exactly the same as for a theorem. The distinction is mainly organizational.
Lemma	Establishes an auxiliary result, often designed for later use in proving a theorem or proposition.	A lemma still requires a complete proof. Its name describes its supporting role, not a lower level of rigor.
Corollary	States a consequence that follows quickly from a preceding result.	Its proof is usually short because most of the work has already been done by the theorem, proposition, or lemma from which it follows.

Two distinctions are especially important. First, **axioms and definitions are starting points, whereas theorems, propositions, lemmas, and corollaries are proved statements**. An axiom supplies an assumption; a definition supplies meaning. Neither is established by proving it from earlier results. Second, among proved statements, the labels describe **expository function rather than mathematical certainty**. A difficult technical lemma may contain the hardest argument in a chapter, while a major theorem may then follow from it in a few lines.

The labels are also somewhat author-dependent. One book may call a useful result a theorem where another calls the same result a proposition. What matters is that you learn to read the local architecture: definitions tell you what words mean; axioms tell you what may be assumed; lemmas build tools; theorems and propositions record established conclusions; corollaries harvest consequences.

Reading habit. When you encounter a labeled statement, ask two questions before reading its proof: *What is being assumed or defined?* and *What earlier result is this statement meant to support or exploit?* This small habit makes the dependency structure of a proof much easier to see.

Course Map

Unit	Topic	Primary proof skill	Counterexample and corrected claim
1	Logic, ordered fields, rational versus real numbers, and proof strategies	Quantifier order, logical negation, contradiction, induction	Converse failures; confusing ordered-field algebra with completeness; no rational square root of 2
2	Completeness, suprema, and the epsilon principle	Translate least-upper-bound logic into approximating witnesses	A bounded rational set may lack a rational supremum; bounded sets need not attain suprema
3	Sequences and limits	Construct an $N(\varepsilon)$ and track dependence	$(-1)^n$; convergence without monotonicity
4	Monotone convergence, subsequences, and Cauchy sequences	Use a supremum first; revisit through nested intervals; analyze tail envelopes	Bounded does not imply convergent
5	Infinite series and rearrangements	Group terms and compare with geometric or p-series models	$a_n \to 0$ does not imply $\sum a_n$ converges
6	Open and closed sets; limit and isolated points	Negate neighborhood definitions; use sequential characterizations	Sets neither open nor closed
7	Compactness, connectedness, and the Cantor set	Extract convergent subsequences; then pass to finite subcovers	Closed but unbounded is not compact
8	Functional limits: Dirichlet and Thomae functions	Translate between sequential and ε–δ criteria	Dirichlet and Thomae functions
9	Continuity and the intermediate value property	Use a supremum first; compare nested-interval and connectedness proofs	Intermediate value without monotonicity
10	Compact domains, Lipschitz continuity, and uniform continuity	Turn point-dependent choices into uniform bounds	x^2 on $\mathbb{R}$; $\sqrt{x}$ is not Lipschitz
11	Derivatives and Darboux's theorem	Rewrite difference quotients; separate existence from regularity	Differentiable with discontinuous derivative
12	Mean value methods, Taylor approximation, and derivative consequences	Construct an auxiliary function; derive first- and second-order estimates	Derivative zero on disconnected domain need not imply constant
13	Riemann integration	Separate high-oscillation intervals of small total length from low-oscillation intervals	One-point modifications; Dirichlet function
14	Integral properties, evaluation and accumulation forms of the FTC, and Lebesgue's criterion	Coordinate partition estimates with local limits	Riemann integrable functions discontinuous on a dense countable set

Logic, Ordered Fields, and Proof Strategies

Suggested Abbott reading: Abbott §§1.1–1.2.
Chapter focus: Logical language, ordered-field structure, the rational and real number systems, and the classical proof that no rational number squares to 2.

Unit at a glance.

Main idea. The language of quantifiers and the order structure of $\mathbb{R}$ are not preliminaries to analysis; they are the grammar in which every later definition is written.

Proof strategy. Translate prose into logic before beginning the algebra.

Counterexample to remember. The principal counterexample and corrected claim are developed in the reflection before the exercises.

Prior knowledge. Sets, functions, divisibility, parity, elementary algebra, and interval notation.

Guiding question

What must be stated explicitly before an algebraic argument counts as a proof?

Inquiry 1.1: Who may depend on whom? *Detachable module*

Investigate. Compare the pairs

$$\forall x \in \mathbb{R}\ \exists y \in \mathbb{R}\ (y > x), \qquad \exists y \in \mathbb{R}\ \forall x \in \mathbb{R}\ (y > x),$$

and

$$\forall \varepsilon > 0\ \exists \delta > 0\ (0 < \delta < \varepsilon), \qquad \exists \delta > 0\ \forall \varepsilon > 0\ (0 < \delta < \varepsilon).$$

Decide which statements are true and propose witnesses when possible.

Decide and justify. For each statement, identify the order in which the variables are chosen, which witness may depend on earlier data, and the exact negation of the statement. Explain why interchanging two quantifiers can change the truth value.

Transfer. Create one new pair of statements of the forms $\forall x\, \exists y$ and $\exists y\, \forall x$ whose truth values differ.

Stop here and record a first response. The resolution is absorbed into the core exposition that follows rather than repeated in a separate answer block.

1.1 Predicates, statements, and domains

Definition 1.1 (Predicate, statement, and domain)**.** An expression such as

$$P(x): \qquad x^2 < 2$$

is a *predicate*: its truth value depends on the value of the free variable x and on the domain from which x is drawn. A variable not governed by a quantifier is *free*; a quantified variable is *bound*. Once every free variable is assigned or quantified, the result is a *statement* with a definite truth value.

For example,

$$\exists x \in \mathbb{R}\ (2x = 1)$$

is true, whereas

$$\exists x \in \mathbb{Z}\ (2x = 1)$$

is false. The formula is the same; the domain changes its truth value.

Throughout, $\mathbb{N} = \{1, 2, 3, \ldots\}$.

The symbols $\forall$ and $\exists$ mean "for every" and "there exists." Before proving a statement, identify

1. the domain of each variable;
2. the order of the quantifiers;
3. which objects are fixed and which are chosen;
4. which later choices may depend on earlier data.

Quantifier order matters. Compare

$$\forall x \in \mathbb{R}\ \exists y \in \mathbb{R}\ (y > x)$$

with

$$\exists y \in \mathbb{R}\ \forall x \in \mathbb{R}\ (y > x).$$

The first is true: after x is given, choose $y = x + 1$. The witness y is allowed to depend on x. The second asks for one fixed real number larger than every real number and is false.

Dependency notation. It is often useful to write $y = y(x)$, $N = N(\varepsilon)$, or $\delta = \delta(\varepsilon, c)$ even when the formal definition does not. This notation records the legal dependencies and prevents accidental use of a choice that depends on data introduced later.

1.2 Negating logical statements

Negation is a mathematical operation, not a change in tone. The basic rules are

$$\neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q), \qquad \neg(P \vee Q) \equiv (\neg P) \wedge (\neg Q),$$

and

$$\neg(P \Rightarrow Q) \equiv P \wedge \neg Q.$$

Thus an implication fails precisely when its hypothesis is true and its conclusion is false.

For quantifiers,

$$\neg(\forall x\ P(x)) \equiv \exists x\ \neg P(x), \qquad \neg(\exists x\ P(x)) \equiv \forall x\ \neg P(x).$$

Each quantifier switches, and the final assertion is negated. The following table contains forms used throughout the course.

Statement	Negation
P and Q	not P or not Q
P or Q	not P and not Q
$P \Rightarrow Q$	P and not Q
$\forall x \in D,\ P(x)$	$\exists x \in D$ such that $\neg P(x)$
$\exists x \in D$ such that $P(x)$	$\forall x \in D,\ \neg P(x)$
$x \leq y$	$x > y$
$\lvert x - a \rvert < \varepsilon$	$\lvert x - a \rvert \geq \varepsilon$

For a nested statement, work from the outside inward:

$$\neg\big(\forall x \in D\ \exists y \in E\ \forall z \in F :\ P(x,y,z)\big)$$

is

$$\exists x \in D\ \forall y \in E\ \exists z \in F : \neg P(x,y,z).$$

In words: there is one x for which every proposed y fails because of some z.

Worked example 1.2 (Boundedness and its negation)**.** A set $A \subseteq \mathbb{R}$ is *bounded above* when

$$\exists M \in \mathbb{R}\ \forall a \in A :\ a \le M.$$

Its negation is

$$\forall M \in \mathbb{R}\ \exists a \in A :\ a > M.$$

Thus "A is not bounded above" means that every proposed upper bound is exceeded by some element of A.

A tempting but false argument. Negating "for every x there exists y" as "for every x there does not exist y." The correct negation is "there exists an x for which no y works."

1.3 Proof grammar

The logical form dictates a natural proof opening.

Logical goal	Standard opening
$\forall x \in D,\ P(x)$	"Let $x \in D$ be arbitrary."
$\exists x \in D,\ P(x)$	"Choose/define $x = \cdots$; we verify $P(x)$."
$P \Rightarrow Q$	"Assume P. We prove Q."
$P \Leftrightarrow Q$	Prove $P \Rightarrow Q$ and $Q \Rightarrow P$ separately.
Uniqueness	Assume two candidates and prove they are equal.
A universal claim is false	Produce one object satisfying the domain but violating the conclusion.

An implication $P \Rightarrow Q$ may be proved directly, by contraposition $\neg Q \Rightarrow \neg P$, or by contradiction: assume P and $\neg Q$ and derive an impossibility. The converse $Q \Rightarrow P$ is a different statement.

Worked example 1.3 (A direct divisibility proof)**.** If m and n are even integers, then $m+n$ is even.

Proof. Let $m, n \in \mathbb{Z}$ be even. By definition, there exist $a, b \in \mathbb{Z}$ such that $m = 2a$ and $n = 2b$. Hence

$$m + n = 2(a+b).$$

Because $a + b \in \mathbb{Z}$, the number $m+n$ is even. □

A tempting but false argument. Writing "let $m = 2a$" without first invoking the definition of evenness. Definitions supply witnesses; the proof must name them and verify that they lie in the required domain.

1.4 The rational and real numbers as ordered fields

Definition 1.4 (Rational and irrational numbers)**.** The *rational numbers* are

$$\mathbb{Q} = \left\{ \frac{m}{n} : m, n \in \mathbb{Z},\ n \neq 0 \right\}.$$

A real number that is not rational is *irrational*. Thus

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}.$$

The first two enlargements repair familiar algebraic closure problems. Passing from $\mathbb{N}$ to $\mathbb{Z}$ makes subtraction possible without leaving the number system, while passing from $\mathbb{Z}$ to $\mathbb{Q}$ makes division by a nonzero integer possible. For example, if $m/n, p/q \in \mathbb{Q}$, then

$$\frac{m}{n} + \frac{p}{q} = \frac{mq + pn}{nq}, \qquad \frac{m}{n}\frac{p}{q} = \frac{mp}{nq},$$

and, when $m \neq 0$, the reciprocal of m/n is n/m. Hence $\mathbb{Q}$ already supports the usual field arithmetic.

Both $\mathbb{Q}$ and $\mathbb{R}$ carry two compatible structures. First, each is a field: addition and multiplication are associative and commutative, multiplication distributes over addition, 0 and 1 are identities, every element has an additive inverse, and every nonzero element has a multiplicative inverse. Second, each has the usual total order. For any two elements x and y of the same system, exactly one of

$$x < y, \qquad x = y, \qquad y < x$$

holds, and the order is compatible with the field operations:

$$\begin{aligned} x < y &\Longrightarrow x + z < y + z, \\ 0 < x,\ 0 < y &\Longrightarrow 0 < xy. \end{aligned}$$

Such a system is called an *ordered field*.

These axioms justify the algebra of inequalities in either $\mathbb{Q}$ or $\mathbb{R}$. For example,

- $x < y$ if and only if $0 < y - x$;
- multiplying an inequality by a positive number preserves its direction;
- multiplying by a negative number reverses its direction;
- $x^2 \geq 0$, with equality only when $x = 0$.

Why squares are nonnegative. If $x = 0$, then $x^2 = 0$. If $x > 0$, then $x^2 > 0$ by compatibility with multiplication. If $x < 0$, then $-x > 0$, so $(-x)^2 > 0$; but $(-x)^2 = x^2$. □

For real numbers, absolute value packages the order into a distance from zero:

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

It satisfies

$$-|x| \leq x \leq |x|, \qquad |x + y| \leq |x| + |y|.$$

For the triangle inequality, add the bounds for x and y to get

$$-(|x| + |y|) \leq x + y \leq |x| + |y|\,,$$

which is exactly $|x + y| \leq |x| + |y|$.

Why enlarge $\mathbb{Q}$ further? The passage from $\mathbb{Q}$ to $\mathbb{R}$ is different from the earlier enlargements: $\mathbb{Q}$ is already an ordered field, so no new rule of ordinary arithmetic is needed. What fails is a completeness property of the number line. Theorem 1.6 exposes a missing rational point; Unit 2 will formulate the precise order property that fills such gaps. We will not construct $\mathbb{R}$ from $\mathbb{Q}$ in this course. Instead, we isolate the structural property that distinguishes the real numbers from the rationals.

1.5 A recurring epsilon argument

Theorem 1.5 (Epsilon principle (course shorthand))**.** *Let $a, b \in \mathbb{R}$. If*

$$a \leq b + \varepsilon \qquad \text{for every } \varepsilon > 0,$$

then $a \leq b$.

Proof plan. The desired conclusion is an order statement. Negate it, use the resulting positive difference to choose an admissible epsilon, and contradict the hypothesis.

Proof. Assume for contradiction that $a > b$, and write $d = a - b > 0$.

Pause and reconstruct: The epsilon principle

Before continuing. Prove that if $a \leq b + \varepsilon$ for every $\varepsilon > 0$, then $a \leq b$.

Guided step. Assume the conclusion fails, so $d = a - b > 0$. Choose $\varepsilon =$ _____. The hypothesis becomes $d \leq$ _____, contradicting $d > 0$. Conclude by returning explicitly to the desired order statement.

Check. Which quantifier licenses an epsilon depending on the positive difference d? Why is the chosen epsilon positive? Where is contradiction used?

Choose $\varepsilon = d/2 = (a - b)/2 > 0$. This is an allowed choice. The hypothesis gives

$$a \leq b + \frac{a - b}{2} = \frac{a + b}{2} < a,$$

a contradiction. Hence $a \leq b$. □

The corresponding zero conclusion is especially useful:

$$0 \leq d < \varepsilon \quad \text{for every } \varepsilon > 0 \qquad \Longrightarrow \qquad d = 0.$$

Consequently, if $|x - y| < \varepsilon$ for every $\varepsilon > 0$, then $x = y$.

Why this is the course motif. The logical vocabulary is completely visible: a universal epsilon hypothesis, the negation of the desired order conclusion, and a witness epsilon chosen from the positive difference. Later proofs repeatedly create an estimate valid for every tolerance and then use the same epsilon argument to force equality, uniqueness, or exact order.

1.6 A gap in $\mathbb{Q}$: no rational square root of 2

Theorem 1.6. *There is no rational number r such that $r^2 = 2$.*

Proof plan. Assume a reduced rational representation $r = p/q$ exists. Use the equation $p^2 = 2q^2$ to force both numerator and denominator to be even, contradicting reduced form.

Proof. Assume for contradiction that $r = p/q \in \mathbb{Q}$ satisfies $r^2 = 2$, where $p, q \in \mathbb{Z}$, $q \neq 0$, and p and q have no common factor. Then

$$p^2 = 2q^2.$$

Thus p^2 is even, so p is even. Write $p = 2k$. Substituting yields

$$4k^2 = 2q^2, \qquad \text{so} \qquad q^2 = 2k^2.$$

Hence q is also even. Then 2 divides both p and q, contradicting the assumption that the fraction is reduced. Therefore no rational number has square 2. □

The supporting fact "if n^2 is even, then n is even" is proved directly by contraposition. If n is odd, then $n = 2k + 1$ and

$$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

is odd.

Return to the opening question. Theorem 1.6 identifies a genuine gap in $\mathbb{Q}$, but it does not yet explain what property of $\mathbb{R}$ fills that gap. Unit 2 turns this example into an order-theoretic question: a bounded set of rational approximations can have a boundary that is missing from $\mathbb{Q}$. The Least Upper Bound Property is the principle that prevents such missing boundaries in $\mathbb{R}$.

1.7 Induction and dependence on earlier cases

We also use the standard *Well-Ordering Principle* for $\mathbb{N}$: every nonempty subset of $\mathbb{N}$ has a least element. This is part of the elementary arithmetic background for the course and will be invoked explicitly when needed.

A statement $P(n)$ for all $n \in \mathbb{N}$ is often proved by establishing a base case and proving $P(n) \Rightarrow P(n+1)$. The inductive hypothesis is available only for the current index n; it is not permission to assume the final result.

Proposition 1.7. *For every $n \in \mathbb{N}$,*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. For $n = 1$, both sides equal 1. Assume the identity holds for some n. Then

$$1 + \cdots + n + (n+1) = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}.$$

Thus the identity holds for $n + 1$, completing the induction. □

Checkpoint. Negate, without using the word "not": "For every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $1/N < \varepsilon$."

Counterexample reflection.

Tempting claim, with context. Every true implication remains true when its hypothesis and conclusion are reversed.

Pause. Write one familiar implication and test its converse on several explicit objects.

Counterexample in this context. If an integer is divisible by 4, then it is even. The converse fails because 6 is even but is not divisible by 4.

Exact failure and repaired claim. The hypothesis was sufficient, not necessary. Keep the forward implication unless the converse receives its own proof.

Formative practice

Exercise 1.1. Fluency. Identify the free variables and domains in each predicate: (a) $x + y > 0$; (b) $n \mid m$; (c) $f(x) = 0$. Then add quantifiers to make one true statement and one false statement from each predicate.

Exercise 1.2. Fluency. Negate each statement symbolically and in words: (a) every real number is rational; (b) there exists a largest real number; (c) every $x > 0$ has some $n \in \mathbb{N}$ with $1/n < x$; (d) $f(x) > 0$ for some $x \in [a, b]$.

Exercise 1.3. Proof practice. Prove from the ordered-field axioms that if $x < y$ and $z < 0$, then $xz > yz$.

Exercise 1.4. Proof practice. Prove that $|x - y| \geq \big||x| - |y|\big|$. Indicate exactly where the triangle inequality is used.

Exercise 1.5. Proof practice. Use the epsilon principle to prove: if $x \geq 0$ and $x \leq c\varepsilon$ for every $\varepsilon > 0$, where $c > 0$ is fixed, then $x = 0$.

Exercise 1.6. Proof practice. Consider the claim: if m^2 is odd, then m is odd. Diagnose two proposed arguments: one proves only that odd m has odd square, while the other proves the contrapositive. State precisely why the first does not prove the claim and why the second does.

Further problems

Exercise 1.7. Strategy choice. Let $f(x) = x^2$, $A = [0, 2]$, and $B = [1, 4]$. Compute $f(A)$ and $f(B)$, test the identities $f(A \cap B) = f(A) \cap f(B)$ and $f(A \cup B) = f(A) \cup f(B)$, and then prove the correct general inclusion or equality for an arbitrary function.

Exercise 1.8. Give a true implication whose converse is false, and explain the failed converse with one explicit object.

Exercise 1.9. Prove by induction that $1 + 3 + \cdots + (2n - 1) = n^2$.

Exercise 1.10. Suppose $r \in \mathbb{Q}$ and $x \notin \mathbb{Q}$. Prove that $r + x$ is irrational. Identify where the closure of $\mathbb{Q}$ under subtraction is used.

What should remain after this unit?
Write a brief answer in your own words.

One statement or distinction I can now formulate precisely:

One proof decision I can reuse:

One counterexample and the claim it repairs:

Completeness, Suprema, and the Epsilon Principle in Action

Suggested Abbott reading: Abbott §§1.3–1.5.
Chapter focus: The Least Upper Bound Property and its consequences: Archimedean behavior, density, nested intervals, and uncountability.

Unit at a glance.

Main idea. The rationals already support ordered-field arithmetic. Completeness is the additional order property that guarantees bounded nonempty subsets of $\mathbb{R}$ have their least upper bounds inside $\mathbb{R}$.

Proof strategy. Use two equivalent descriptions of a supremum: least among all upper bounds, or an upper bound with points of the set arbitrarily close from below.

Counterexample to remember. The principal counterexample and corrected claim are developed in the reflection before the exercises.

Prior knowledge. Unit 1 logic, the ordered-field structure of $\mathbb{Q}$ and $\mathbb{R}$, absolute value, and set notation.

Guiding question

If $\mathbb{Q}$ already supports the usual ordered-field arithmetic, what is missing from $\mathbb{Q}$, and how does completeness repair it in $\mathbb{R}$?

Opening discussion

Why enlarge $\mathbb{Q}$ to $\mathbb{R}$ if addition, subtraction, multiplication, division, and order already work in $\mathbb{Q}$?

Theorem 1.6 shows that no rational number has square 2. The issue is therefore not a failure of ordinary algebra. The missing feature is an order boundary: rational numbers can lie on one side of the equation $x^2 = 2$ without any rational number occupying the boundary itself. This unit makes that defect precise through least upper bounds and then states the property of $\mathbb{R}$ that rules it out.

Pacing options for this unit.

Core. Completeness, upper bounds, suprema, the epsilon characterization, and the Archimedean argument.

Second pass. Density and the Nested Interval Property as reusable consequences of completeness.

Enrichment. Uncountability and the second proof through nested intervals.

Applied-analysis connection: certified bisection

Suppose a continuous model changes sign across an interval. Bisection repeatedly halves the interval and produces nested closed intervals. Completeness guarantees that the nested intervals have a common real point in $\mathbb{R}$; the same existence conclusion need not hold in an incomplete ordered field. The later Intermediate Value Theorem uses continuity to identify that point as a root. *Analytical question:* which step guarantees existence of a candidate point, and which later hypothesis determines its function value?

Inquiry 2.1: Supremum witness workshop *Detachable module*

Investigate. Let

$$A = \{1 - 1/n : n \in \mathbb{N}\}.$$

Before using any theorem, decide whether 1 is an upper bound, a maximum, the supremum, or more than one of these.

Decide and justify. Separate the two obligations needed to prove $\sup A = 1$: first prove that 1 is an upper bound; then, given $\varepsilon > 0$, choose n so that $1 - \varepsilon < 1 - 1/n$. Analyze the statement "1 is the supremum because it is the largest element."

Transfer. Repeat the two-part verification for $(0, 2)$. Which words in the proof change, and which proof strategy remains identical?

Stop here and record a first response. The resolution is absorbed into the core exposition that follows rather than repeated in a separate answer block.

2.1 From $\mathbb{Q}$ to $\mathbb{R}$: the missing-boundary problem

Unit 1 showed two facts that should be kept separate. The rationals $\mathbb{Q}$ form an ordered field, so the usual arithmetic and inequality rules already work there. Nevertheless, no rational number has square 2.

Consider the rational set

$$A_{\mathbb{Q}} = \left\{q \in \mathbb{Q} : q \geq 0,\ q^2 < 2\right\}.$$

It is nonempty because $1 \in A_{\mathbb{Q}}$, and it is bounded above in $\mathbb{Q}$; for example, 2 is an upper bound. The natural boundary between the conditions $q^2 < 2$ and $q^2 > 2$ is precisely where $q^2 = 2$ would hold, but Theorem 1.6 shows that no rational number occupies that boundary. We now introduce the language that makes this failure precise.

We are not constructing $\mathbb{R}$ from $\mathbb{Q}$. Rather, we use this rational gap to identify the additional order property that the real numbers possess.

2.2 Upper bounds and suprema: the logic written out

Definition 2.1 (Upper and lower bounds; bounded sets)**.** Let F be an ordered field and let $A \subseteq F$. A number $u \in F$ is an *upper bound for A in F* when

$$\forall a \in A :\ a \leq u,$$

and $\ell \in F$ is a *lower bound for A in F* when

$$\forall a \in A :\ \ell \leq a.$$

The set A is *bounded above* if it has an upper bound, *bounded below* if it has a lower bound, and *bounded* if both conditions hold. When the ambient field is $\mathbb{R}$, we usually omit the words "in $\mathbb{R}$."

Definition 2.2 (Supremum and infimum)**.** Let $A \subseteq F$, where F is an ordered field. A number $s \in F$ is the *supremum of A in F* when

1. s is an upper bound:
$$\forall a \in A :\ a \leq s;$$
2. s is no larger than any upper bound in F:
$$\forall u \in F, \qquad \big[(\forall a \in A :\ a \leq u) \Rightarrow s \leq u\big].$$

Thus the supremum is the least upper bound. Dually, $r \in F$ is the *infimum of A in F* if r is a lower bound and every lower bound ℓ satisfies $\ell \le r$; it is the greatest lower bound. When the ambient field matters, we write $\sup_F A$ and $\inf_F A$. For subsets of $\mathbb{R}$, we normally omit the subscript.

Definition 2.3 (Maximum and minimum)**.** A number m is the *maximum* of A if $m \in A$ and $a \le m$ for every $a \in A$; dually, m is the *minimum* if $m \in A$ and $m \le a$ for every $a \in A$. We write $\max A$ and $\min A$ when these extrema exist. A supremum or infimum need not belong to the set.

The next calculation is the common algebraic engine for both the rational gap and the real completeness argument. Isolating it once keeps the distinction between *ordered-field algebra* and *completeness* visible.

Lemma 2.4 (Boundary adjustment near $x^2 = 2$)**.** *For $r \ge 1$, define*

$$T(r) = \frac{2(r+1)}{r+2}.$$

Then

$$T(r) - r = \frac{2 - r^2}{r+2}, \qquad 2 - T(r)^2 = \frac{2(2-r^2)}{(r+2)^2}.$$

Consequently,

- *if $r^2 < 2$, then $T(r) > r$ and $T(r)^2 < 2$;*
- *if $r^2 > 2$, then $0 < T(r) < r$ and $T(r)^2 > 2$.*
- *If $r \in \mathbb{Q}$, then $T(r) \in \mathbb{Q}$.*

Proof. Direct subtraction gives

$$T(r) - r = \frac{2(r+1) - r(r+2)}{r+2} = \frac{2 - r^2}{r+2},$$

and direct squaring gives

$$2 - T(r)^2 = 2 - \frac{4(r+1)^2}{(r+2)^2} = \frac{2(2-r^2)}{(r+2)^2}.$$

Because $r \ge 1$, the denominators are positive and $T(r) > 0$. The two sign conclusions follow immediately. Finally, the displayed formula for $T(r)$ uses only rational field operations, so rational input gives rational output. □

Proposition 2.5 (The rational gap at 2)**.** *The set*

$$A_{\mathbb{Q}} = \{q \in \mathbb{Q} : q \ge 0,\ q^2 < 2\}$$

is nonempty and bounded above in $\mathbb{Q}$, but $\sup_{\mathbb{Q}} A_{\mathbb{Q}}$ does not exist.

Proof. We already observed that $1 \in A_{\mathbb{Q}}$ and that 2 is an upper bound. Suppose, for contradiction, that $s = \sup_{\mathbb{Q}} A_{\mathbb{Q}}$. Then $s \ge 1$, and Theorem 1.6 gives $s^2 \ne 2$.

If $s^2 < 2$, Lemma 2.4 gives a rational number $T(s) > s$ with $T(s)^2 < 2$. Thus $T(s) \in A_{\mathbb{Q}}$, contradicting that s is an upper bound.

If $s^2 > 2$, the same lemma gives $0 < T(s) < s$ and $T(s)^2 > 2$. Every $q \in A_{\mathbb{Q}}$ must satisfy $q < T(s)$: otherwise $q \ge T(s) > 0$ would imply $q^2 \ge T(s)^2 > 2$, contradicting $q \in A_{\mathbb{Q}}$. Hence $T(s)$ is an upper bound for $A_{\mathbb{Q}}$ smaller than s, contradicting the leastness of s.

Both cases are impossible, so no rational supremum exists. □

Axiom 2.6 (Least Upper Bound Property)**.** *Every nonempty subset of* $\mathbb{R}$ *that is bounded above has a supremum in* $\mathbb{R}$*.*

This is the completeness axiom for the real numbers used throughout the course. We do not prove it from the ordered-field axioms; rather, it is the additional structural assumption that distinguishes $\mathbb{R}$ from $\mathbb{Q}$ for the purposes of this course. Applying the axiom to $-A = \{-a : a \in A\}$ also gives the dual fact that every nonempty set bounded below has an infimum.

Proposition 2.7 (Completeness fills the gap)**.** *Let*

$$A_{\mathbb{R}} = \left\{x \in \mathbb{R} : x \geq 0,\ x^2 < 2\right\}.$$

Then $A_{\mathbb{R}}$ *has a supremum* $s \in \mathbb{R}$*, and* $s^2 = 2$*. This unique positive number is denoted by* $\sqrt{2}$*. In particular, Theorem 1.6 shows that* $\sqrt{2} \notin \mathbb{Q}$*.*

Proof. The set $A_{\mathbb{R}}$ is nonempty because $1 \in A_{\mathbb{R}}$ and is bounded above by 2. By Axiom 2.6, $s = \sup A_{\mathbb{R}}$ exists, with $s \geq 1$.

If $s^2 < 2$, Lemma 2.4 gives $T(s) > s$ and $T(s)^2 < 2$. Hence $T(s) \in A_{\mathbb{R}}$, contradicting that s is an upper bound.

If $s^2 > 2$, the same lemma gives $0 < T(s) < s$ and $T(s)^2 > 2$. If $x \in A_{\mathbb{R}}$ and $x \geq T(s)$, then $x > 0$ and $x^2 \geq T(s)^2 > 2$, contradicting $x \in A_{\mathbb{R}}$. Thus every $x \in A_{\mathbb{R}}$ satisfies $x < T(s)$, so $T(s)$ would be an upper bound smaller than s, contradicting leastness. Therefore $s^2 = 2$.

Finally, a positive solution is unique: if $x, y > 0$ and $x^2 = y^2 = 2$, then $(x - y)(x + y) = 0$, and $x + y > 0$, so $x = y$. □

What completeness adds. The same boundary-adjustment calculation works in both $\mathbb{Q}$ and $\mathbb{R}$. In $\mathbb{Q}$, it proves that a least upper bound for $A_{\mathbb{Q}}$ cannot exist. In $\mathbb{R}$, the Least Upper Bound Property first supplies a candidate s, and the calculation then forces $s^2 = 2$. The algebra is shared; the existence of the boundary point is the extra content of completeness.

Worked example 2.8. For $A = (0, 1)$, we have $\sup A = 1$, but $1 \notin A$. The number 1 is the least upper bound, not a maximum.

2.3 The epsilon characterization of the supremum

The leastness condition in the definition is exact, but it is often awkward to verify directly. The following equivalent form replaces comparison with every upper bound by one upper-bound check and one approximation condition.

Lemma 2.9 (Epsilon characterization of the supremum)**.** *Let* $A \subseteq \mathbb{R}$ *be nonempty and let* $s \in \mathbb{R}$*. Then* $s = \sup A$ *if and only if both of the following hold:*

(1) s *is an upper bound for* A*:*

$$\forall a \in A :\ a \leq s;$$

(2) elements of A *occur arbitrarily close to* s *from below:*

$$\forall \varepsilon > 0\ \exists a_\varepsilon \in A \qquad \textit{such that} \qquad s - \varepsilon < a_\varepsilon.$$

Proof. Suppose first that $s = \sup A$. By definition, s is an upper bound. Let $\varepsilon > 0$. If no element of A satisfied $s - \varepsilon < a$, then every $a \in A$ would satisfy $a \le s - \varepsilon$. Thus $s - \varepsilon$ would be an upper bound smaller than s, contradicting the leastness of s. Hence a suitable $a_\varepsilon \in A$ exists.

Conversely, suppose conditions (1) and (2) hold. Let u be any upper bound for A. If $u < s$, choose $\varepsilon = s - u > 0$. Condition (2) gives $a_\varepsilon \in A$ with

$$a_\varepsilon > s - \varepsilon = u,$$

contradicting that u is an upper bound. Hence $s \le u$ for every upper bound u. Together with (1), this says exactly that s is the least upper bound of A. □

The forward implication uses the negation

$$\neg\big(\exists a \in A :\ s - \varepsilon < a\big) \quad \Longleftrightarrow \quad \forall a \in A :\ a \le s - \varepsilon,$$

which is the Unit 1 quantifier rule in action.

Proof plan. To prove $s = \sup A$, use a two-part ledger: (i) prove $a \le s$ for every $a \in A$; (ii) given arbitrary $\varepsilon > 0$, construct an element $a_\varepsilon \in A$ with $s - \varepsilon < a_\varepsilon$. Lemma 2.9 guarantees that these two obligations are not merely sufficient; together they are equivalent to $s = \sup A$.

2.4 Archimedean consequences

Theorem 2.10 (Archimedean Property I)**.** *For every $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $n > x$.*

Proof. If $\mathbb{N}$ were bounded above, Axiom 2.6 would give $s = \sup \mathbb{N}$. By Lemma 2.9 with $\varepsilon = 1$, there would exist $n \in \mathbb{N}$ with $s - 1 < n \le s$. Then $n + 1 \in \mathbb{N}$ and $n + 1 > s$, contradicting that s is an upper bound. □

A useful reciprocal form of the Archimedean property is

Corollary 2.11 (Archimedean Property II)**.**

$$\forall \varepsilon > 0\ \exists n \in \mathbb{N} :\ \frac{1}{n} < \varepsilon.$$

Proof. In Theorem 2.10, replace x by $1/\varepsilon$. Then note that since $\varepsilon > 0$, $n > 1/\varepsilon$ implies $1/n < \varepsilon$. □

Corollary 2.12 (Discrete epsilon principle)**.** *If $a, b \in \mathbb{R}$ satisfy*

$$a \le b + \frac{1}{n} \qquad \text{for every } n \in \mathbb{N},$$

then $a \le b$.

Proof. If $a > b$, then $a - b > 0$. By Corollary 2.11, we can choose n with $1/n < a - b$. The assumed inequality gives $a - b \le 1/n < a - b$, a contradiction. □

In particular,

$$0 \le d \le \frac{1}{n} \quad \text{for every } n \in \mathbb{N} \qquad \Longrightarrow \qquad d = 0.$$

This is Theorem 1.5 combined with Theorem 2.10: the values $1/n$ supply arbitrarily small positive tolerances.

Lemma 2.13 (Integer-part step)**.** *For every $x \in \mathbb{R}$, there exists $m \in \mathbb{Z}$ such that*

$$x < m \leq x + 1.$$

Proof. By Theorem 2.10, choose $k \in \mathbb{N}$ with $k > -x$. By the Well-Ordering Principle stated in Unit 1, every nonempty subset of $\mathbb{N}$ has a least element. The set

$$S = \{j \in \mathbb{N} : j > x + k\}$$

is nonempty by Theorem 2.10 applied to $x + k$. Let j_0 be its least element and set $m = j_0 - k$. We have $m > x$. If $j_0 > 1$, minimality gives $j_0 - 1 \leq x + k$; if $j_0 = 1$, the same inequality follows from $x + k > 0$. Hence $m \leq x + 1$. □

Theorem 2.14 (Density of the rationals)**.** *If $a < b$, then there exists $r \in \mathbb{Q}$ with $a < r < b$.*

Proof plan. Choose n so large that the grid spacing $1/n$ is smaller than $b - a$. Then use the integer-part lemma to choose a grid point just to the right of na.

Proof. By Theorem 2.10, choose $n \in \mathbb{N}$ such that $1/n < b - a$. Apply Lemma 2.13 to na to obtain $m \in \mathbb{Z}$ with

$$na < m \leq na + 1.$$

Then $a < m/n \leq a + 1/n < b$, so $r = m/n$ is rational and lies between a and b. □

The irrationals are dense as well. Given $a < b$, Proposition 2.7 constructs $\sqrt{2}$, and Theorem 1.6 shows it is irrational. By Theorem 2.14, choose a rational r in $(a - \sqrt{2}, b - \sqrt{2})$ and set $x = r + \sqrt{2}$. Then $a < x < b$. Moreover, x cannot be rational: otherwise $\sqrt{2} = x - r$ would be rational because $\mathbb{Q}$ is a field, contradicting Theorem 1.6.

2.5 Nested intervals

The notation $I_1, I_2, I_3, \ldots$ below denotes an indexed family of intervals. No notion of sequence convergence is needed for this theorem.

Theorem 2.15 (Nested Interval Property)**.** *Let $I_n = [a_n, b_n]$ be nonempty closed intervals with*

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots.$$

Then $\bigcap_{n=1}^{\infty} I_n$ is nonempty.

Proof. The set $A = \{a_n : n \in \mathbb{N}\}$ is nonempty. It is bounded above because $I_n \subseteq I_1$ for every n, so $a_n \leq b_1$; hence b_1 is an upper bound. By Axiom 2.6, let $x = \sup A$.

Pause and reconstruct: Nested intervals from completeness

Before continuing. Let $I_n = [a_n, b_n]$ be nested nonempty closed intervals. Reconstruct the proof that their intersection is nonempty.

Guided step. Set $A = \{a_n : n \in \mathbb{N}\}$ and $x = \sup A$. Complete the two inequalities $a_n \le x$ and $x \le b_n$. For the second, show that each b_n is an upper bound for every a_k by splitting into the cases $k \ge n$ and $k < n$.

Check. Where is completeness used? Where is nestedness used? Why does proving both inequalities place x in every interval?

For each n, $a_n \le x$ because $a_n \in A$. We claim that b_n is an upper bound for all a_k. If $k \ge n$, then $I_k \subseteq I_n$, so $a_k \in I_n$ and hence $a_k \le b_n$. If $k < n$, then $I_n \subseteq I_k$, so $a_k \le a_n \le b_n$. Thus b_n is an upper bound for A, and the least-upper-bound property gives $x \le b_n$. Therefore

$$a_n \le x \le b_n$$

for every n, so $x \in I_n$ for every n. Hence $x \in \bigcap_{n=1}^{\infty} I_n$. □

2.6 Uncountability of $\mathbb{R}$

Definition 2.16 (Countable and countably infinite sets)**.** A set A is *countably infinite* if there exists a bijection $f : \mathbb{N} \to A$. A set is *countable* if it is finite or countably infinite.

The integers are countable, as witnessed by the listing

$$0, 1, -1, 2, -2, 3, -3, \ldots.$$

The rationals are countable as well: list the pairs $(m, n) \in \mathbb{Z} \times \mathbb{N}$ by increasing $|m| + n$, send (m, n) to m/n, and delete repeated rational values. We now show that $(0, 1)$, and hence $\mathbb{R}$, cannot be listed in this way.

Theorem 2.17 (Cantor)**.** *The interval* $(0, 1)$ *is uncountable.*

Proof plan. First use Cantor's diagonal argument: change one selected digit in each proposed entry so that the constructed number differs from every listed number. Then revisit the theorem through nested intervals to display a reusable completeness argument.

Proof (via Cantor's diagonal argument). Suppose, for contradiction, that

$$(0, 1) = \{x_1, x_2, x_3, \ldots\}.$$

Write each x_n in a decimal expansion that is not eventually all 9's,

$$x_1 = 0.d_{11}d_{12}d_{13}\ldots, \quad x_2 = 0.d_{21}d_{22}d_{23}\ldots, \quad \ldots.$$

Choose $y = 0.c_1c_2c_3\ldots$ with $c_n = 1$ when $d_{nn} \ne 1$ and $c_n = 2$ when $d_{nn} = 1$. Then $y \in (0, 1)$ and differs from x_n in the nth digit. The digits 1 and 2 avoid the ambiguity of terminating decimal expansions. Thus y is absent from the proposed list, a contradiction. □

Proof (via the nested interval property). Suppose again that $(0, 1) = \{x_1, x_2, x_3, \ldots\}$. Choose a nonempty closed interval $I_1 \subset (0, 1)$ with $x_1 \notin I_1$. Once I_n has been chosen, divide it into three equal closed subintervals and choose an outer third that does not contain x_{n+1}; call it I_{n+1}. Then

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots, \qquad x_n \notin I_n.$$

Theorem 2.15 supplies $x \in \bigcap_n I_n$. Since $x \in I_n$ but $x_n \notin I_n$, the point x is absent from the list. □

Connection. Cantor's diagonal argument is the shorter of the two proofs. However, the proof using the nested interval property is also quite effective here. We will see more applications of the nested intervals in the proofs of: monotone convergence, and the Intermediate Value Theorem, in later units.

A tempting but false argument. From $s = \sup A$, concluding $s \in A$. This is true only when the supremum is attained.

Checkpoint. Let $A = \{x \in \mathbb{R} : x^2 < 5\}$. Explain how completeness and the epsilon characterization in Lemma 2.9 can be used to define and characterize $\sqrt{5}$ as $\sup(A \cap [0, \infty))$.

Counterexample reflection.

Tempting claim, with context. Every nonempty subset of $\mathbb{R}$ that is bounded above has a maximum.

Pause. Compare maximum with supremum. Which notion requires membership in the set?

Counterexample in this context. The set $(0, 1)$ has supremum 1, but $1 \notin (0, 1)$, so it has no maximum.

Exact failure and repaired claim. Completeness guarantees a least upper bound, not attainment. Replace 'maximum' by 'supremum.'

Return to the opening question. Completeness does not assert that a bounded set contains its supremum. It asserts that every nonempty set bounded above has a least upper bound in $\mathbb{R}$; Lemma 2.9 then says that elements of the set occur arbitrarily close to that supremum from below. The corresponding rational set for $\sqrt{2}$ shows what can fail in an incomplete ordered field.

Formative practice

Exercise 2.1. Fluency. Write in quantifiers: (a) u is an upper bound for A; (b) A is bounded above; (c) A has a maximum; (d) A has no maximum. Then negate (a)–(c).

Exercise 2.2. Fluency. Find the supremum and infimum of $(0, 1)$, $[0, 1)$, and $\{1 - 1/n : n \in \mathbb{N}\}$. State whether each is attained.

Exercise 2.3. Proof practice. Let $A \subseteq \mathbb{R}$ be nonempty and let $r \in \mathbb{R}$. Prove that $r = \inf A$ if and only if (i) $r \leq a$ for every $a \in A$, and (ii) for every $\varepsilon > 0$ there exists $a_\varepsilon \in A$ such that $a_\varepsilon < r + \varepsilon$.

Exercise 2.4. Proof practice. If $s = \sup A$ and $c > 0$, prove $cs = \sup(cA)$ where $cA = \{ca : a \in A\}$. Separate the upper-bound and epsilon-approximation steps.

Exercise 2.5. Proof practice. Use the discrete epsilon principle to prove: if $x \leq y + 2/n$ for every $n \in \mathbb{N}$, then $x \leq y$.

Exercise 2.6. Proof practice. Prove that between any two distinct real numbers there is an irrational number.

Further problems

Exercise 2.7. Let A be nonempty and bounded below, and let B be the set of all lower bounds of A. Prove that $\sup B = \inf A$. Explain why the Least Upper Bound Property also supplies greatest lower bounds.

Exercise 2.8. For nonempty bounded-above sets $A, B \subseteq \mathbb{R}$, define $A + B = \{a + b : a \in A,\ b \in B\}$. Prove

$$\sup(A + B) = \sup A + \sup B.$$

Separate the upper-bound argument from the epsilon-approximation argument.

Exercise 2.9. Let

$$A_n = \{a_{n1}, a_{n2}, a_{n3}, \ldots\}, \qquad n \in \mathbb{N},$$

be nonempty countable sets, each supplied with an enumeration; repetitions are allowed when a set is finite. Arrange the enumerations as the rows of the infinite array

$$\begin{array}{ccccc} a_{11} & a_{12} & a_{13} & a_{14} & \cdots \\ a_{21} & a_{22} & a_{23} & a_{24} & \cdots \\ a_{31} & a_{32} & a_{33} & a_{34} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{array}.$$

Read the entries by successive diagonals:

$$a_{11}; \qquad a_{12}, a_{21}; \qquad a_{13}, a_{22}, a_{31}; \qquad \ldots.$$

Explain why every element of $\bigcup_{n=1}^{\infty} A_n$ eventually appears in this diagonal enumeration. If an element appears more than once, keep only its first occurrence. Conclude that the union is countable. Finally, explain how this diagonal *enumeration* differs from Cantor's diagonal argument above.

What should remain after this unit?
Write a brief answer in your own words.

One statement or distinction I can now formulate precisely:

One proof decision I can reuse:

One counterexample and the claim it repairs:

Sequences and the ε–N Definition

Suggested Abbott reading: Abbott §§2.2–2.3.
Chapter focus: Limits of sequences, uniqueness, algebraic limit laws, order arguments, and proof construction from an ε target.

Unit at a glance.

Main idea. A sequence is a function from $\mathbb{N}$ to $\mathbb{R}$, usually written (a_n). It converges when its tail lies inside every prescribed neighborhood of one fixed number.

Proof strategy. Work backward from $|a_n - L| < \varepsilon$ to a sufficient condition on n.

Counterexample to remember. The principal counterexample and corrected claim are developed in the reflection before the exercises.

Prior knowledge. Absolute values, inequalities, elementary algebra, and the Nested Interval Property from Unit 2.

Guiding question

What mathematical object is a sequence, and what does it mean, with exact quantifier order, for its terms to be eventually close to one number?

3.1 Sequences and convergence

Analysis repeatedly studies an infinite list of approximations, measurements, or iterates. Before asking whether such a list converges, we should specify what the object itself is.

Definition 3.1 (Sequence)**.** A *sequence of real numbers* is a function

$$a : \mathbb{N} \to \mathbb{R}.$$

We write $a_n = a(n)$ and denote the sequence by (a_n). The index is part of the data: order matters and repetitions are allowed, so a sequence is not merely the set of values that occur. The sequence is *bounded* if its range $\{a_n : n \in \mathbb{N}\}$ is a bounded subset of $\mathbb{R}$; equivalently, there exists $M > 0$ such that $|a_n| \leq M$ for every n. For $N \in \mathbb{N}$, the terms $a_N, a_{N+1}, \ldots$ form a *tail* of the sequence.

The central question is now meaningful: when should the terms a_n be regarded as approaching one fixed real number L? The answer must control every sufficiently late term, not merely infinitely many selected terms.

Inquiry 3.1: Constructing a response to ε *Detachable module*

Investigate. For

$$c_n = \frac{5n^2 - 4n}{2n^2 + 7},$$

predict the limit and decide whether the sequence lies above or below it. After subtracting the proposed limit, identify the numerator and denominator of the absolute error.

Decide and justify. Use the following bounding pattern: place a simple *upper bound* on the error numerator and a simple *lower bound* on the positive denominator. Convert the result to C/n. For $\varepsilon = 10^{-1}$ and $\varepsilon = 10^{-2}$, produce valid integer choices of N; optimality is not required.

Transfer. Before reading the worked example, decide which inequalities justify replacing $8n + 35$ by a multiple of n and $4n^2 + 14$ by a multiple of n^2. Explain why the directions of these replacements matter.

Stop here and record a first response. The resolution is absorbed into the core exposition that follows rather than repeated in a separate answer block.

Mathematics refresh. For nonnegative quantities, increasing the numerator or decreasing the positive denominator makes a fraction larger. Thus, if $0 \le A \le B$ and $C \ge D > 0$, then $A/C \le B/D$. This is the order principle behind the rational-sequence estimates.

Review pointer: Review absolute values, fraction arithmetic, and the rule that division by a positive number preserves inequalities.

Definition 3.2 (Convergence of a sequence)**.** A sequence (a_n) *converges to* L if

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ \forall n \ge N, \qquad |a_n - L| < \varepsilon.$$

We write $a_n \to L$.

Worked example 3.3. The sequence $a_n = 1/n$ converges to 0.

Proof. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ with $N > 1/\varepsilon$. If $n \ge N$, then

$$|a_n - 0| = \frac{1}{n} \le \frac{1}{N} < \varepsilon.$$

Thus $a_n \to 0$. □

Proposition 3.4 (Shrinking nested intervals)**.** *Let $I_n = [a_n, b_n]$ be nonempty closed intervals satisfying*

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots.$$

If $b_n - a_n \to 0$, then

$$\bigcap_{n=1}^{\infty} I_n$$

contains exactly one point.

Proof. Theorem 2.15 gives at least one point in the intersection. To prove uniqueness, suppose x and y both lie in every I_n. Then

$$0 \le |x - y| \le b_n - a_n \qquad \text{for every } n.$$

Given $\varepsilon > 0$, convergence of $b_n - a_n$ to 0 gives $N \in \mathbb{N}$ such that $b_n - a_n < \varepsilon$ for every $n \ge N$. In particular,

$$0 \le |x - y| \le b_N - a_N < \varepsilon.$$

Because this holds for every $\varepsilon > 0$, Theorem 1.5 gives $|x - y| = 0$, and hence $x = y$. □

Connection. Unit 2 used completeness to obtain *existence* of a common point for arbitrary nested nonempty closed intervals. The new convergence hypothesis $b_n - a_n \to 0$ upgrades that conclusion to *uniqueness*. This is why the shrinking version belongs naturally in the sequence unit.

Worked example 3.5 (A rational sequence: bound the numerator above and the denominator below)**.** Consider

$$c_n = \frac{5n^2 - 4n}{2n^2 + 7}.$$

The leading terms suggest the candidate limit $5/2$. We now prove this directly and record which side of the limit contains the sequence.

Proof. Let $\varepsilon > 0$. Subtracting the candidate limit gives

$$c_n - \frac{5}{2} = \frac{2(5n^2 - 4n) - 5(2n^2 + 7)}{2(2n^2 + 7)} = -\frac{8n + 35}{4n^2 + 14} < 0.$$

Thus $c_n < 5/2$. For $n \geq 1$,

$$8n + 35 \leq 43n, \qquad 4n^2 + 14 \geq 4n^2,$$

so

$$\left| c_n - \frac{5}{2} \right| = \frac{8n + 35}{4n^2 + 14} \leq \frac{43n}{4n^2} = \frac{43}{4n}.$$

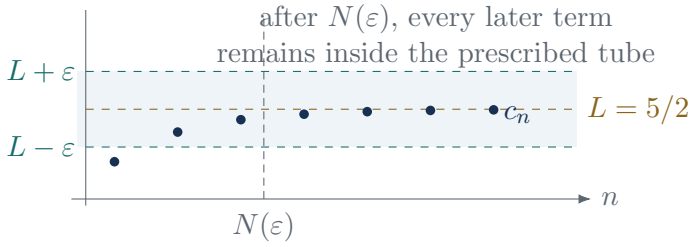
Choose any $N \in \mathbb{N}$ with $N > 43/(4\varepsilon)$. If $n \geq N$, then

$$\left| c_n - \frac{5}{2} \right| \leq \frac{43}{4n} \leq \frac{43}{4N} < \varepsilon.$$

Therefore $c_n \to 5/2$, and the sequence approaches its limit from below. □

The estimation template. For a positive fraction, an upper bound on the numerator and a lower bound on the denominator produce an upper bound on the whole fraction. Exact simplification is useful, but the proof only needs a convenient estimate tending to zero.

Connection. A valid cutoff need not be optimal: bound the error by C/n and choose any integer $N > C/\varepsilon$.



The quantifier order is visible: after the tolerance fixes the tube around $L = 5/2$, one chooses a cutoff $N(\varepsilon)$; every later term must remain inside. Here the terms approach the limit from below.

Proof plan. The proof begins with an arbitrary $\varepsilon > 0$. Solve, or safely overestimate, the target inequality for n. Then choose an integer $N = N(\varepsilon)$ large enough to make the sufficient condition hold.

The negation of convergence to L is

$$\exists \varepsilon_0 > 0 \ \forall N \in \mathbb{N} \ \exists n \geq N \quad \text{such that} \quad |a_n - L| \geq \varepsilon_0.$$

Thus nonconvergence means that some fixed neighborhood is escaped infinitely far out in the sequence.

3.2 Uniqueness and boundedness

Theorem 3.6 (Uniqueness of limits)**.** *If $a_n \to L$ and $a_n \to M$, then $L = M$.*

Checkpoint. Suppose $L \neq M$. Which tolerance, chosen from $|L - M|$, makes it impossible for one sufficiently late term to lie close to both L and M?

Proof. Assume $L \neq M$ and set $\varepsilon = |L - M| / 3$. Because both limits control a tail of the same sequence, there is one index N such that for every $n \geq N$,

$$|a_n - L| < \varepsilon, \qquad |a_n - M| < \varepsilon.$$

For such an n, the triangle inequality gives

$$|L - M| \leq |L - a_n| + |a_n - M| < 2\varepsilon = \frac{2}{3} |L - M|,$$

a contradiction. Hence $L = M$. □

Theorem 3.7. *Every convergent sequence is bounded.*

Proof. If $a_n \to L$, choose N so that $|a_n - L| < 1$ for $n \geq N$. Then $|a_n| \leq |L| + 1$ on the tail. The finitely many initial terms also have a finite maximum absolute value. Taking the larger of these numbers bounds the whole sequence. □

3.3 Algebraic and order limit laws

The next rules turn convergence estimates into an algebra. Suppose throughout that

$$a_n \to a, \qquad b_n \to b, \qquad c \in \mathbb{R}.$$

Then

$$a_n + b_n \to a + b, \qquad ca_n \to ca, \qquad a_n b_n \to ab.$$

If $b \neq 0$ and the quotients a_n / b_n are defined for all sufficiently large n, then

$$\frac{a_n}{b_n} \to \frac{a}{b}.$$

Subtraction is included by taking the scalar $c = -1$.

Guided proof: the sum law. Start from

$$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b|.$$

How small should each error be to make their sum less than ε? After obtaining two cutoff indices, which single cutoff works for both tails?

Sum. Let $\varepsilon > 0$. Choose $N_1, N_2 \in \mathbb{N}$ such that

$$n \geq N_1 \Longrightarrow |a_n - a| < \frac{\varepsilon}{2}, \qquad n \geq N_2 \Longrightarrow |b_n - b| < \frac{\varepsilon}{2}.$$

With $N = \max\{N_1, N_2\}$, every $n \geq N$ satisfies

$$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b| < \varepsilon.$$

Thus $a_n + b_n \to a + b$.

Guided proof: scalar multiplication. If $c \neq 0$, the identity

$$|ca_n - ca| = |c|\ |a_n - a|$$

shows exactly how the requested output tolerance must be rescaled. What should replace ε in the convergence estimate for $a_n \to a$? Treat $c = 0$ separately.

Scalar multiplication. If $c = 0$, the conclusion is immediate. If $c \neq 0$, let $\varepsilon > 0$ and choose N so that

$$n \geq N \Longrightarrow |a_n - a| < \frac{\varepsilon}{|c|}.$$

Then for $n \geq N$,

$$|ca_n - ca| = |c|\ |a_n - a| < \varepsilon.$$

Hence $ca_n \to ca$.

Guided proof: the product law. The useful decomposition is

$$a_n b_n - ab = a_n(b_n - b) + b(a_n - a).$$

The second term is controlled directly by $a_n \to a$. The first contains the extra factor a_n. Which theorem from the preceding section supplies a fixed bound for that factor?

Product. By Theorem 3.7, (a_n) is bounded. Choose $M \geq 1$ such that $|a_n| \leq M$ for every n. Given $\varepsilon > 0$, choose N_1, N_2 so that

$$n \geq N_1 \Longrightarrow |b_n - b| < \frac{\varepsilon}{2M}, \qquad n \geq N_2 \Longrightarrow |a_n - a| < \frac{\varepsilon}{2(|b| + 1)}.$$

For $n \geq N = \max\{N_1, N_2\}$,

$$\begin{aligned} |a_n b_n - ab| &= |a_n(b_n - b) + b(a_n - a)| \\ &\leq |a_n|\,|b_n - b| + |b|\,|a_n - a| \\ &< M\frac{\varepsilon}{2M} + |b|\,\frac{\varepsilon}{2(|b| + 1)} < \varepsilon. \end{aligned}$$

Therefore $a_n b_n \to ab$.

Guided proof: the quotient law. Before estimating a reciprocal, first keep the denominator away from zero. Since $b \neq 0$, require

$$|b_n - b| < \frac{|b|}{2}.$$

Use the reverse triangle inequality to obtain a positive lower bound for $|b_n|$. Then estimate $|1/b_n - 1/b|$ and finish with the product law.

Reciprocal and quotient. Because $b_n \to b \neq 0$, there is N_0 such that for $n \geq N_0$,

$$|b_n - b| < \frac{|b|}{2}.$$

The reverse triangle inequality then gives

$$|b_n| \geq |b| - |b_n - b| > \frac{|b|}{2},$$

so in particular $b_n \neq 0$ eventually. Let $\varepsilon > 0$, and choose N_1 so that

$$n \geq N_1 \Longrightarrow |b_n - b| < \frac{\varepsilon\,|b|^2}{2}.$$

For $n \geq \max\{N_0, N_1\}$,

$$\left|\frac{1}{b_n} - \frac{1}{b}\right| = \frac{|b_n - b|}{|b_n|\,|b|} \leq \frac{2}{|b|^2}\,|b_n - b| < \varepsilon.$$

Thus $1/b_n \to 1/b$. The product law now yields

$$\frac{a_n}{b_n} = a_n \frac{1}{b_n} \longrightarrow a\frac{1}{b} = \frac{a}{b}.$$

Because convergence is a tail property, finitely many earlier indices play no role.

Four reusable proof moves. The algebraic laws introduce four patterns that recur throughout analysis: split an error into simpler pieces; rescale the tolerance; bound an extra factor uniformly; and keep a denominator away from zero before taking reciprocals.

Hypothesis check. The condition $b \neq 0$ in the quotient law is essential. For $b_n = 1/n$, we have $b_n \to 0$, but $1/b_n = n$ does not converge.

Theorem 3.8 (Order Limit Theorem)**.** *If $a_n \leq b_n$ for all sufficiently large n, $a_n \to a$, and $b_n \to b$, then $a \leq b$.*

Proof. If $a > b$, let $\varepsilon = (a - b)/3$. Eventually $a_n > a - \varepsilon$ and $b_n < b + \varepsilon$. But

$$a - \varepsilon > b + \varepsilon,$$

so eventually $a_n > b_n$, contradicting the assumed order. □

Corollary 3.9 (Squeeze Theorem)**.** *If $a_n \leq x_n \leq b_n$ eventually and $a_n, b_n \to L$, then $x_n \to L$.*

Proof. Fix $\varepsilon > 0$. For all sufficiently large n, both a_n and b_n lie in $(L - \varepsilon, L + \varepsilon)$, and the eventual inequalities $a_n \leq x_n \leq b_n$ then force $L - \varepsilon < x_n < L + \varepsilon$. Hence $x_n \to L$. □

A tempting but false argument. Choosing $N = 1/\varepsilon$ without ensuring that N is an integer, or letting N depend on n. The dependency is $N = N(\varepsilon)$, followed by all $n \geq N$.

Checkpoint. Prove directly that $(3n + 1)/(n + 4) \to 3$ by estimating the difference from 3.

Counterexample reflection.

Tempting claim, with context. Every bounded sequence converges.

Pause. Construct a sequence that remains in a small set but has two incompatible tail behaviors.

Counterexample in this context. The sequence $a_n = (-1)^n$ is bounded, while its even and odd subsequences have different limits.

Exact failure and repaired claim. Boundedness prevents escape but not oscillation. Add monotonicity, or weaken the conclusion to existence of a convergent subsequence.

Formative practice

Exercise 3.1. Fluency. Prove from the definition that $5/n \to 0$.

Exercise 3.2. Proof practice. Prove $1/\sqrt{n} \to 0$. Make your choice of N explicit.

Exercise 3.3. Proof practice. Prove directly that $(8n+3)/(5n-4) \to 8/5$. After subtracting $8/5$, upper-bound the numerator and lower-bound the denominator; do not solve the exact inequality unless needed.

Exercise 3.4. Proof practice. If $a_n \to a$, prove $|a_n| \to |a|$.

Exercise 3.5. Proof practice. Suppose $a_n \geq 0$ and $a_n \to a$. Prove $a \geq 0$.

Exercise 3.6. Fluency. A sequence is *eventually* in A if every sufficiently late term lies in A, and *frequently* in A if arbitrarily late terms lie in A. Compare these notions for $(-1)^n$ and $A = \{1\}$, then restate convergence using "eventually."

Further problems

Exercise 3.7. Reverse the first two quantifiers in the definition of convergence: assume there exists $\varepsilon > 0$ such that for every N and every $n \geq N$, $|x_n - x| < \varepsilon$. Characterize exactly which sequences satisfy this altered condition and give one divergent example. The invented word "verconges" may be used only inside this exercise.

Exercise 3.8. Let $a_n \to a$, and let p and q be polynomials with $q(a) \neq 0$. Use the algebraic limit laws to prove

$$\frac{p(a_n)}{q(a_n)} \longrightarrow \frac{p(a)}{q(a)}.$$

Explain where the condition $q(a) \neq 0$ enters the argument.

Exercise 3.9. Let (z_n) be the shuffled sequence $(x_1, y_1, x_2, y_2, \ldots)$. Prove that (z_n) converges if and only if (x_n) and (y_n) converge to the same limit.

What should remain after this unit?
Write a brief answer in your own words.

One statement or distinction I can now formulate precisely:

One proof decision I can reuse:

One counterexample and the claim it repairs:

Monotone Convergence, Subsequences, and Cauchy Sequences

Suggested Abbott reading: Abbott §§2.4–2.6.
Chapter focus: Monotone convergence, subsequences, Bolzano–Weierstrass, and the Cauchy Criterion.

Unit at a glance.

Main idea. Boundedness alone does not force convergence, but it forces a convergent subsequence; the Cauchy property characterizes convergence in $\mathbb{R}$.

Proof strategy. Relate whole sequences to selected tails or subsequences.

Counterexample to remember. The principal counterexample and corrected claim are developed in the reflection before the exercises.

Prior knowledge. Suprema, nested intervals, and ε–N convergence.

Guiding question

What structure can still be extracted from a bounded sequence that does not itself converge?

Inquiry 4.1: What does boundedness actually buy? *Detachable module*

Investigate. Compare

$$(-1)^n, \qquad 1-\frac{1}{n}, \qquad \frac{(-1)^n}{n}.$$

For each sequence, decide whether it is bounded, monotone, convergent, and whether sufficiently late terms become arbitrarily close to one another. Then identify all visible subsequential limits.

Decide and justify. Separate three possible conclusions: convergence of the whole sequence, convergence of at least one subsequence, and small variation within every sufficiently late tail. Decide which additional structure—monotonicity, subsequence extraction, or late-term closeness—supports each conclusion.

Transfer. Construct a bounded sequence with exactly two subsequential limits. Then state the repaired theorem that boundedness supports in place of convergence of the whole sequence.

Stop here and record a first response. The resolution is absorbed into the core exposition that follows rather than repeated in a separate answer block.

Applied-analysis connection: stopping criteria and tail estimates

In an iterative computation, bounded output alone does not certify convergence: the iterates may oscillate. A monotonicity argument, a contraction estimate, or a bound on distances between late iterates supplies a convergence estimate. *Analytical question:* does the available estimate bound distance from a proposed limit, or only distances between late iterates?

4.1 Monotone convergence

Definition 4.1 (Monotone sequence)**.** A sequence is *increasing* if $a_n \le a_{n+1}$ for every n, and *decreasing* if $a_{n+1} \le a_n$ for every n.

Theorem 4.2 (Monotone Convergence Theorem)**.** *Every increasing sequence that is bounded above converges. Every decreasing sequence that is bounded below converges.*

Proof plan. Use the following three-step proof plan: (1) Axiom 2.6 gives $L = \sup\{a_n : n \in \mathbb{N}\}$; (2) Lemma 2.9 gives $a_N > L - \varepsilon$; (3) monotonicity traps every later term between $L - \varepsilon$ and L.

Supremum proof. Let (a_n) be increasing and bounded above. By Axiom 2.6, the nonempty set $\{a_n : n \in \mathbb{N}\}$ has a least upper bound; set

$$L = \sup\{a_n : n \in \mathbb{N}\}.$$

Given $\varepsilon > 0$, the number $L - \varepsilon$ is not an upper bound. Hence some N satisfies $a_N > L - \varepsilon$. For every $n \geq N$, monotonicity and the upper-bound property give

$$L - \varepsilon < a_N \leq a_n \leq L.$$

Thus $|a_n - L| < \varepsilon$, so $a_n \to L$. For a decreasing sequence bounded below, apply the increasing case just proved to $(-a_n)$. □

4.1.1 A second proof through nested intervals

Choose an upper bound M and begin with $I_1 = [a_1, M]$. Bisect I_1. If the entire sequence lies in the left half, retain it; otherwise some term lies in the right half, and monotonicity places the entire later tail there. Repeat. This constructs nested closed intervals I_k whose lengths tend to zero, each containing an entire tail of the sequence.

Proposition 3.4 supplies the unique point $L \in \bigcap_k I_k$. Given $\varepsilon > 0$, choose k with $\text{length}(I_k) < \varepsilon$. The retained tail and L lie in the same interval I_k, so all sufficiently late terms satisfy $|a_n - L| < \varepsilon$.

Connection. The supremum proof is the one to use first. The nested-interval proof is a second proof: it shows how nested intervals repeatedly reduce the possible location of the limit until uniqueness follows.

Worked example family: Heron's iteration for $\sqrt{2}$ as monotone convergence

Proposition 2.7 gives the positive real number $\sqrt{2}$. Start with $x_0 \geq \sqrt{2}$ and define

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right).$$

The identity

$$x_{n+1} - \sqrt{2} = \frac{(x_n - \sqrt{2})^2}{2x_n} \geq 0$$

keeps the sequence above $\sqrt{2}$, while

$$x_n - x_{n+1} = \frac{x_n^2 - 2}{2x_n} \geq 0$$

shows that it is decreasing. Theorem 4.2 gives a limit $L \geq \sqrt{2} > 0$, and the recurrence yields $L = (L + 2/L)/2$, hence $L^2 = 2$. By the uniqueness statement in Proposition 2.7, $L = \sqrt{2}$. This is Heron's method; it is also Newton's method applied to $x^2 - 2 = 0$. The point of the example is not only the formula but the proof obligations: invariant bound, monotonicity, existence of the limit, and identification of the limit.

Hypothesis check. Both hypotheses—increasing and bounded above—in the Monotone Convergence Theorem are necessary. The sequence n is increasing but unbounded. The sequence $(-1)^n$ is bounded but not monotone.

4.2 Subsequences

Definition 4.3 (Subsequence)**.** A *subsequence* of (a_n) has the form (a_{n_k}), where

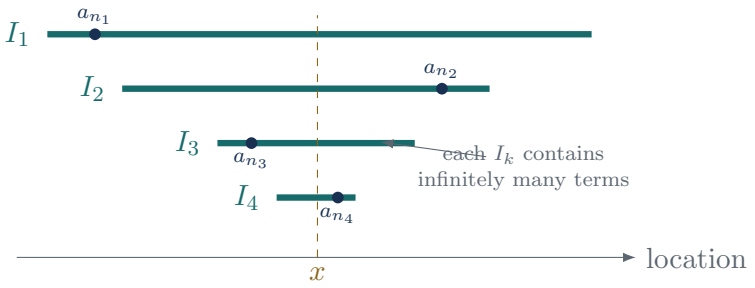
$$n_1 < n_2 < n_3 < \cdots .$$

If $a_n \to L$, then every subsequence also converges to L.

Theorem 4.4 (Bolzano–Weierstrass)**.** *Every bounded sequence in* $\mathbb{R}$ *has a convergent subsequence.*

Proof plan. Repeatedly bisect a bounded interval and retain a half containing infinitely many terms. The nested intervals shrink to one point; choose one new sequence term from each retained interval.

Proof. Place the bounded sequence in a closed interval I_1. Bisect I_1; at least one half contains infinitely many terms. Call that half I_2. Continue, producing nested closed intervals I_k with length tending to zero, each containing infinitely many terms. Proposition 3.4 shows that their intersection is a single point x.



Successive bisection retains one interval containing infinitely many terms. That condition permits a fresh choice $n_k > n_{k-1}$ with $a_{n_k} \in I_k$; the shrinking intervals then force the selected subsequence toward their common point x.

Choose $n_k > n_{k-1}$ with $a_{n_k} \in I_k$. Since a_{n_k} and x lie in I_k,

$$|a_{n_k} - x| \le \text{length}(I_k) \to 0.$$

Thus $a_{n_k} \to x$. □

A useful divergence test follows: if a sequence has two subsequences converging to different limits, then the original sequence diverges.

4.3 Extended exercise: tail envelopes, $\limsup$ and $\liminf$

Definition 4.5 (Tail envelopes, limit superior, and limit inferior)**.** For a bounded sequence (a_n), define

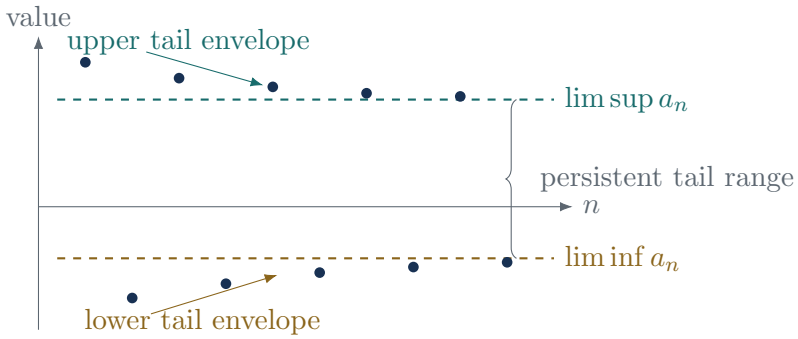
$$M_n = \sup\{a_k : k \ge n\}, \qquad m_n = \inf\{a_k : k \ge n\}.$$

The sequences (M_n) and (m_n) are the *upper* and *lower tail envelopes*. The sequence (M_n) is decreasing and bounded below, while (m_n) is increasing and bounded above. Therefore Theorem 4.2 guarantees that both limits below exist. We define

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} M_n, \qquad \liminf_{n\to\infty} a_n = \lim_{n\to\infty} m_n.$$

Operational consequence of Definition 4.5. If $L = \limsup a_n$ and $q > L$, then $a_n < q$ eventually; if $q < L$, then $a_n > q$ infinitely often. Indeed, $M_n \to L$, so eventually $M_n < q$ in the first case, while eventual $a_n \le q$ in the second case would force $M_n \le q$ eventually. The corresponding statements for $\liminf$ follow by reversing the inequalities.

As the first terms are discarded, M_n can only move downward and m_n can only move upward. Later, for nonnegative sequences that may be unbounded, we use the same tail formulas in the extended sense: an unbounded tail has supremum $+\infty$, and a monotone tail-envelope limit may equal $+\infty$. This is all that is needed for the ratio and root tests.



Exercise 4.1 (Extended)**.** Work through the following steps.

(a) Prove that (M_n) is decreasing and (m_n) is increasing. Explain why both are bounded.

(b) Deduce from the Monotone Convergence Theorem that the two displayed limits exist.

(c) Prove $m_n \le a_n \le M_n$ and hence $\liminf a_n \le \limsup a_n$.

(d) Show that a_n converges if and only if $\liminf a_n = \limsup a_n$, in which case both equal $\lim a_n$.

(e) Compute the two values for $a_n = (-1)^n + 1/n$.

(f) Show that every subsequential limit lies between $\liminf a_n$ and $\limsup a_n$. Then show that each endpoint is itself a subsequential limit.

Picture to retain. $\limsup a_n$ is the eventual upper level repeatedly approached by the tail; $\liminf a_n$ is the corresponding lower level. They ignore finitely many early terms and measure persistent oscillation.

4.4 Cauchy sequences

Why the Cauchy condition? Convergence compares late terms with a fixed candidate limit L. In many approximation procedures, however, the limiting value is not known in advance. We therefore ask whether the late terms are close to one another, using only information internal to the sequence. This leads to the Cauchy condition.

Definition 4.6 (Cauchy sequence)**.** A sequence is *Cauchy* if

$$\forall \varepsilon > 0 \ \exists N \ \forall m > n \ge N, \qquad |a_m - a_n| < \varepsilon.$$

The definition detects stabilization without naming a candidate limit. The Cauchy Criterion below says that completeness of $\mathbb{R}$ turns this internal tail control into convergence to an actual real number.

Theorem 4.7. *Every convergent sequence is Cauchy.*

Proof. If $a_n \to L$, choose N so that $|a_n - L| < \varepsilon/2$ for $n \geq N$. Then for $m, n \geq N$,

$$|a_m - a_n| \leq |a_m - L| + |a_n - L| < \varepsilon.$$ □

Theorem 4.8 (Cauchy Criterion)**.** *A sequence of real numbers converges if and only if it is Cauchy.*

Proof. Only the converse remains. Every Cauchy sequence is bounded: choose a tail whose terms stay within 1 of one fixed tail term, then absorb the finite initial segment. By Theorem 4.4, it has a subsequence $a_{n_k} \to L$. Given $\varepsilon > 0$, choose N_1 so that $m, n \geq N_1$ implies $|a_m - a_n| < \varepsilon/2$. Choose k with $n_k \geq N_1$ and $|a_{n_k} - L| < \varepsilon/2$. Then for $n \geq N_1$,

$$|a_n - L| \leq |a_n - a_{n_k}| + |a_{n_k} - L| < \varepsilon.$$

Thus $a_n \to L$. □

Connection. The Cauchy Criterion is another form of completeness. In $\mathbb{Q}$, a rational sequence can be Cauchy while approaching an irrational number, so it need not converge within $\mathbb{Q}$.

Checkpoint. Find all subsequential limits of $a_n = (-1)^n + 1/n$.

Counterexample reflection.

Tempting claim, with context. Every Cauchy sequence of rational numbers converges to a rational number.

Pause. Separate the internal tail estimate from completeness of the ambient number system.

Counterexample in this context. Rational decimal truncations of $\sqrt{2}$ form a Cauchy sequence in $\mathbb{Q}$, but their real limit is not rational.

Exact failure and repaired claim. Cauchy behavior alone does not create a limit inside an incomplete space. Work in $\mathbb{R}$ or separately prove that the limit belongs to $\mathbb{Q}$.

Formative practice

Exercise 4.2. Proof practice. Prove that the recursively defined sequence $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$ converges. Identify a bound and prove monotonicity.

Exercise 4.3. Strategy choice. Let $x_0 = 2$ and $x_{n+1} = \frac{1}{2}(x_n + 2/x_n)$. Prove that $x_n \geq \sqrt{2}$, that (x_n) is decreasing, and that its limit is $\sqrt{2}$. Then replace 2 by $c > 0$.

Exercise 4.4. Fluency. Give an explicit subsequence of $(-1)^n$ converging to 1 and one converging to -1.

Exercise 4.5. Proof practice. Prove directly that every Cauchy sequence is bounded.

Further problems

Exercise 4.6. Determine all subsequential limits of $a_n = \cos(n\pi/2) + 1/n$. Then decide whether the sequence is Cauchy, justifying the answer from its late tails rather than merely citing nonconvergence.

Exercise 4.7. Prove that if every convergent subsequence of a bounded sequence has the same limit L, then the whole sequence converges to L.

Exercise 4.8. Give a Cauchy sequence in $\mathbb{Q}$ that does not converge in $\mathbb{Q}$. You may use decimal truncations of $\sqrt{2}$.

Exercise 4.9. Suppose (a_n) and (b_n) are Cauchy. Prove directly, without invoking convergence or the Algebraic Limit Laws, that $(a_n + b_n)$ and $(|a_n - b_n|)$ are Cauchy. Then decide whether the converse implication for $(|a_n - b_n|)$ is valid.

Synthesis checkpoint: what should remain?
Complete this panel in your own words. There is no single prescribed set of answers.

A definition or distinction I can now state:

A theorem or claim I want to retain:

A proof strategy I can reuse:

A counterexample and the claim it repairs:

A question I still have:

Infinite Series: Model Families and Comparison

Suggested Abbott reading: Abbott §§2.1, 2.4, 2.7–2.8.
Chapter focus: Series as limits of partial sums; geometric series; grouped p-series estimates; comparison, limit comparison, ratio, root, and alternating-series tests; absolute convergence and rearrangement.

Unit at a glance.

Main idea. A convergence test is not a trick chosen from a list. It is a justified comparison with a series whose behavior is already understood.

Proof strategy. Reduce an unfamiliar series to one of two model families: geometric series or p-series.

Counterexample to remember. The principal counterexample and corrected claim are developed in the reflection before the exercises.

Prior knowledge. Sequences, monotone convergence, Cauchy estimates, tail envelopes, and powers.

Guiding question

How can finite partial sums, strategic grouping, and two model families organize the theory of infinite sums?

Pacing options for this unit.

Core. Partial sums, geometric series, p-series, direct comparison, limit comparison, and the alternating-series argument.

Second pass. Ratio and root tests as geometric comparison in disguise; absolute convergence as a transfer principle.

Enrichment. Rearrangements and the distinction between absolute and conditional convergence.

Opening discussion

Why can grouping or rearranging infinitely many terms appear to change a sum, although grouping finitely many terms never does?

The phrase "add forever" conceals the real object: a sequence of finite partial sums. Once that sequence is visible, grouping becomes a comparison of selected partial sums and rearrangement becomes a question about how much mass can be moved. Geometric series and p-series will serve as the two model families.

Inquiry 5.1: Choose the model before the test *Detachable module*

Investigate. For each series, first inspect signs, the term limit, and dominant scale:

$$\sum \frac{3i+1}{i^2+5}, \qquad \sum \frac{i}{3^i}, \qquad \sum \frac{(-1)^{i+1}}{\sqrt{i}}, \qquad \sum \frac{1}{i^2+4}.$$

Decide and justify. Before calculating, choose a comparison target or proof strategy: harmonic or p-series behavior, geometric decay, ratio/root detection, or alternating cancellation. Then predict convergence, divergence, and—when relevant—absolute versus conditional convergence.

Transfer. Explain why the Ratio and Root Tests cannot distinguish $\sum 1/i$ from $\sum 1/i^2$ when both tests return the inconclusive value 1, and identify the model family that does distinguish them.

Stop here and record a first response. The resolution is absorbed into the core exposition that follows rather than repeated in a separate answer block.

Mathematics refresh. For $r \neq 1$,

$$1 + r + \cdots + r^{n-1} = \frac{1-r^n}{1-r}.$$

When $|r| < 1$, the term r^n tends to zero. The infinite geometric sum is therefore a limit of finite identities, not an instruction to "add forever."

Review pointer: Review factoring the difference $(1-r)(1+r+\cdots+r^{n-1}) = 1-r^n$ and the behavior of powers r^n.

5.1 What an infinite sum means

A finite sum is produced by finitely many additions. An infinite series cannot be defined by "adding forever." Given a sequence (a_i), define its *partial sums* by

$$S_n = \sum_{i=1}^{n} a_i.$$

The index i labels the summands, while n labels the partial sums. We keep this distinction throughout the unit.

Definition 5.1 (Convergence of a series)**.** The series $\sum_{i=1}^{\infty} a_i$ *converges to* S when the partial-sum sequence (S_n) converges to S. Thus every question about an infinite series is first a question about the sequence of its finite partial sums.

Proposition 5.2 (Term Test)**.** *If $\sum a_i$ converges, then $a_i \to 0$.*

Proof. If $S_n \to S$, then the shifted partial sums S_{n-1} also converge to S. For $n \geq 2$, the difference $S_n - S_{n-1}$ is exactly the nth summand. The algebraic limit laws therefore show that the summands tend to $S - S = 0$. Hence $a_i \to 0$ as $i \to \infty$. □

Theorem 5.3 (Cauchy Criterion for Series)**.** *The series $\sum a_i$ converges if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $m > n \geq N$,*

$$|a_{n+1} + a_{n+2} + \cdots + a_m| < \varepsilon.$$

Proof. Apply Theorem 4.8 to the partial-sum sequence (S_n). If $m > n$, then

$$S_m - S_n = a_{n+1} + a_{n+2} + \cdots + a_m.$$ □

In words: a series converges precisely when every sufficiently late finite block has arbitrarily small total contribution.

5.2 The first model: geometric series

For real numbers a and r, take the geometric summands

$$a_i = ar^{i-1}, \qquad i \in \mathbb{N}.$$

Then the nth partial sum is

$$S_n = \sum_{i=1}^{n} a_i = a + ar + ar^2 + \cdots + ar^{n-1}.$$

Multiplication by r shifts every term:

$$rS_n = ar + ar^2 + \cdots + ar^n.$$

Subtracting gives

$$(1 - r)S_n = a - ar^n,$$

so, when $r \neq 1$,

$$S_n = a\frac{1 - r^n}{1 - r}.$$

If $|r| < 1$, then $r^n \to 0$ and

$$\sum_{i=1}^{\infty} ar^{i-1} = \frac{a}{1 - r}.$$

If $|r| \geq 1$ and $a \neq 0$, the terms fail to tend to zero. A nontrivial geometric series therefore converges exactly when $|r| < 1$.

5.3 Positive series and direct comparison

For $a_i \geq 0$, the partial sums increase. By Theorem 4.2, the series therefore converges exactly when those partial sums are bounded above.

Theorem 5.4 (Comparison Test)**.** *Suppose $a_i, b_i \geq 0$ eventually.*

(a) If $a_i \leq b_i$ eventually and $\sum b_i$ converges, then $\sum a_i$ converges.

(b) If $b_i \leq a_i$ eventually and $\sum b_i$ diverges, then $\sum a_i$ diverges.

Proof. In (a), the increasing partial sums of $\sum a_i$ are eventually bounded by a fixed constant coming from $\sum b_i$. Part (b) is the contrapositive of (a), with the roles reversed. □

A tempting but false argument. The comparison must point in the useful direction. Being smaller than a divergent series proves nothing; being larger than a convergent series proves nothing.

5.4 The second model: p-series by writing and grouping terms

We simply write consecutive summands in groups of sizes

$$1, 2, 4, 8, \dots$$

and compare each parenthesized group with one simple numerical bound.

5.4.1 First picture: $p = 1$

Write the harmonic series so that, after the first term, the blocks have sizes $1, 2, 4, 8, \dots$:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \left(\frac{1}{9} + \cdots + \frac{1}{16}\right) + \cdots .$$

The one-term block $1/2$ equals $1/2$. In the next block, both terms are at least $1/4$, so the block is at least $2(1/4) = 1/2$. In the following block, all four terms are at least $1/8$, so the block is at least $4(1/8) = 1/2$. Continuing in the same way, every displayed block contributes at least $1/2$. Thus

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots \geq 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots .$$

The grouped partial sums grow without bound, so the harmonic series diverges.

5.4.2 Second picture: $p = 2$

Use the same group sizes:

$$1 + \left(\frac{1}{2^2} + \frac{1}{3^2}\right) + \left(\frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2}\right) + \left(\frac{1}{8^2} + \cdots + \frac{1}{15^2}\right) + \cdots .$$

Now replace every term in a group by its first, and therefore largest, term:

$$\sum_{i=1}^{\infty} \frac{1}{i^2} \leq 1 + 2\frac{1}{2^2} + 4\frac{1}{4^2} + 8\frac{1}{8^2} + \cdots = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots .$$

The right side is geometric, so $\sum 1/i^2$ converges.

5.4.3 The same picture for $p > 1$

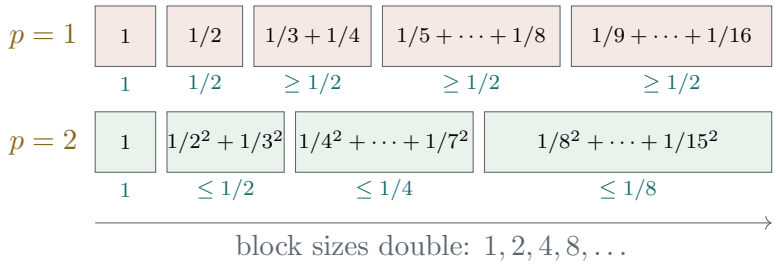
Again write groups of sizes $1, 2, 4, 8, \dots$:

$$1 + \left(\frac{1}{2^p} + \frac{1}{3^p}\right) + \left(\frac{1}{4^p} + \cdots + \frac{1}{7^p}\right) + \left(\frac{1}{8^p} + \cdots + \frac{1}{15^p}\right) + \cdots .$$

The same replacement gives

$$\sum_{i=1}^{\infty} \frac{1}{i^p} \leq 1 + \frac{2}{2^p} + \frac{4}{4^p} + \frac{8}{8^p} + \cdots = 1 + 2^{1-p} + 4^{1-p} + 8^{1-p} + \cdots .$$

This is a geometric series with ratio $2^{1-p} < 1$. Hence the p-series converges for $p > 1$. For $0 < p \leq 1$, we have $1/i^p \geq 1/i$, so the harmonic series gives divergence; for $p \leq 0$, the terms do not tend to zero.



Dyadic grouping changes the scale of inspection. For $p = 1$, every new block retains at least $1/2$ of mass. For $p = 2$, the block masses are bounded by a convergent geometric series.

Worked example 5.5 (The same grouping beyond p-series)**.** Consider

$$\sum_{i=2}^{\infty} \frac{1}{i \log i}.$$

The groups beginning at $2, 4, 8, 16, \ldots$ look like

$$\left(\frac{1}{2\log 2} + \frac{1}{3\log 3}\right) + \left(\frac{1}{4\log 4} + \cdots + \frac{1}{7\log 7}\right) + \left(\frac{1}{8\log 8} + \cdots + \frac{1}{15\log 15}\right) + \cdots.$$

In the group beginning at 2^k, there are 2^k terms. Each is at least

$$\frac{1}{2^{k+1}(k+1)\log 2},$$

so the whole group is at least $1/[2(k+1)\log 2]$. The grouped totals therefore dominate a constant multiple of

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots,$$

and the original series diverges. The proof step is concrete: write the terms, group them, and estimate one entire group at a time.

Connection. Grouping preserves every term but changes the scale at which we inspect accumulation. The harmonic series has terms tending to zero, yet each doubled block retains a fixed amount of mass.

5.5 Limit comparison as the main transfer test

Exact inequalities can be awkward. Usually the real question is whether two positive sequences have the same eventual scale.

Theorem 5.6 (Limit Comparison Test)**.** *Let $a_i, b_i > 0$ eventually, and suppose*

$$\lim_{i\to\infty} \frac{a_i}{b_i} = L, \qquad 0 < L < \infty.$$

Then $\sum a_i$ and $\sum b_i$ either both converge or both diverge.

Pause and reconstruct: Limit comparison

Before continuing. Assume $a_i, b_i > 0$ eventually and $a_i/b_i \to L$ with $0 < L < \infty$. Prove that the two positive series have the same convergence behavior.

Guided step. Use $\varepsilon = L/2$ to obtain

$$_____\, b_i < a_i < _____\, b_i$$

eventually. Apply direct comparison once to transfer convergence and once, in the opposite direction, to transfer divergence.

Check. Where are positivity and the condition $0 < L < \infty$ used? Why do finitely many exceptional terms not affect convergence?

Proof. With $\varepsilon = L/2$, eventually

$$\frac{L}{2} < \frac{a_i}{b_i} < \frac{3L}{2}.$$

Thus a_i is trapped between two positive constant multiples of b_i, and direct comparison applies in both directions. □

Constant factors do not affect convergence. Rational, algebraic, and logarithmic expressions should usually be tested first against a p-series. For example,

$$\frac{3i+7}{i^3-1} \sim \frac{3}{i^2},$$

so the associated series converges by limit comparison with $\sum 1/i^2$.

Proof plan. Identify the dominant behavior. If a_i resembles C/i^p, use a p-series. If it resembles Cq^i, use a geometric series. Limit comparison is the default transfer tool when the quotient approaches a positive finite constant.

5.6 Ratio and root tests: geometric comparison in disguise

The next tests detect geometric behavior. The tail-envelope language from Unit 4 lets the tests work even when the relevant ratios or roots do not have ordinary limits.

Theorem 5.7 (Ratio Test)**.** *Assume $a_i \neq 0$ eventually.*

(a) If

$$\limsup_{i\to\infty} \left|\frac{a_{i+1}}{a_i}\right| < 1,$$

then $\sum a_i$ converges absolutely.

(b) If

$$\liminf_{i\to\infty} \left|\frac{a_{i+1}}{a_i}\right| > 1,$$

then $\sum a_i$ diverges.

At the value 1, the test is inconclusive.

Proof. For (a), choose $q < 1$ strictly above the limsup. Eventually $|a_{i+1}| \leq q\,|a_i|$, and iteration bounds the tail by a geometric series. For (b), choose $q > 1$ strictly below the liminf. Eventually $|a_{i+1}| \geq q\,|a_i|$, so the terms cannot tend to zero. □

Theorem 5.8 (Root Test)**.** *Let*

$$L = \limsup_{i\to\infty} \sqrt[i]{|a_i|}.$$

If $L < 1$, then $\sum a_i$ converges absolutely. If $L > 1$, then $\sum a_i$ diverges. When $L = 1$, the test is inconclusive.

Proof. If $L < 1$, choose q with $L < q < 1$. Eventually $\sqrt[i]{|a_i|} < q$, hence $|a_i| < q^i$. If $L > 1$, choose q with $1 < q < L$. Infinitely many i satisfy $\sqrt[i]{|a_i|} > q$, so $|a_i| > q^i$ along a subsequence and in particular $a_i \not\to 0$. □

5.7 Absolute convergence and alternating series

Definition 5.9 (Absolute and conditional convergence)**.** A series is *absolutely convergent* if $\sum |a_i|$ converges. A convergent series that is not absolutely convergent is *conditionally convergent*.

Absolute convergence implies convergence because, whenever $m > n \geq N$,

$$|a_{n+1} + \cdots + a_m| \leq |a_{n+1}| + \cdots + |a_m|,$$

and the right-hand absolute tail is small for sufficiently large N. Conditional convergence must instead come from cancellation.

Theorem 5.10 (Alternating Series Test)**.** *If $b_i \geq 0$, $b_{i+1} \leq b_i$, and $b_i \to 0$, then*

$$\sum_{i=1}^{\infty} (-1)^{i+1} b_i$$

converges.

Proof. The even partial sums increase and the odd partial sums decrease; every even partial sum lies below every odd one. Theorem 4.2 gives limits for both subsequences. The gap between an odd partial sum and the preceding even partial sum is one summand, whose magnitude tends to zero because $b_i \to 0$. The algebraic limit laws of Section 3.3 therefore force the two subsequential limits to agree. □

As a first example separating two notions, the alternating harmonic series converges but not absolutely, and is therefore conditionally convergent.

Hypothesis check. The hypotheses "decreasing" and "tends to zero" do different jobs. Decrease traps the two partial-sum subsequences; convergence to zero forces the difference between them to converge to zero.

5.8 Rearrangements

A *rearrangement* changes the order of the terms but neither deletes nor duplicates any of them. Formally, if $\pi : \mathbb{N} \to \mathbb{N}$ is a bijection, then

$$\sum_{i=1}^{\infty} a_{\pi(i)}$$

is a rearrangement of $\sum_{i=1}^{\infty} a_i$. Its partial sums will be denoted

$$T_n = \sum_{i=1}^{n} a_{\pi(i)}.$$

The central question is whether changing the order changes the limiting behavior of (T_n).

Orientation. Every rearrangement of an absolutely convergent real series converges to the same sum. The proof, which we do not need later, captures a sufficiently long finite initial block and uses the small absolute tail to control whatever terms a rearrangement moves across that block. Conditional convergence behaves differently: cancellation can depend on order. The examples below make both phenomena visible.

Worked example 5.11 (A harmless rearrangement of an absolutely convergent series)**.** Start from

$$\sum_{i=1}^{\infty} \frac{1}{2^i} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = 1.$$

Swap each adjacent pair:

$$\frac{1}{4} + \frac{1}{2} + \frac{1}{16} + \frac{1}{8} + \frac{1}{64} + \frac{1}{32} + \cdots .$$

Let (T_n) be the partial sums of this rearranged series, and set $S_0 = 0$ for convenience. After every complete pair, nothing has changed:

$$T_{2n} = S_{2n}.$$

Hence $T_{2n} \to 1$. Between two completed pairs,

$$T_{2n-1} = S_{2n-2} + \frac{1}{2^{2n}},$$

and the added term tends to zero while $S_{2n-2} \to 1$. Thus $T_{2n-1} \to 1$ as well. The even and odd partial sums have the same limit, so

$$T_n \to 1.$$

This example does not prove the general rearrangement theorem, but it shows concretely why moving absolutely summable mass is harmless.

Worked example 5.12 (A conditional rearrangement that changes the sum)**.** Consider the alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots .$$

The Alternating Series Test shows that it converges. Its odd partial sums decrease, and

$$S_3 = 1 - \frac{1}{2} + \frac{1}{3} = \frac{5}{6},$$

so its original sum L satisfies $L \leq 5/6 < 1$.

Now separate the positive and negative terms:

$$1 + \frac{1}{3} + \frac{1}{5} + \cdots , \qquad -\frac{1}{2} - \frac{1}{4} - \frac{1}{6} - \cdots .$$

The positive subseries diverges to $+\infty$ because

$$\frac{1}{2j-1} \geq \frac{1}{2j},$$

while the magnitudes of the negative terms form

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \cdots = \frac{1}{2} \sum_{j=1}^{\infty} \frac{1}{j},$$

which also diverges.

We can therefore build a rearrangement as follows. Take unused positive terms, in their original order, until the running partial sum first exceeds 1. Then take unused negative terms until it first falls below 1. Repeat this procedure indefinitely. Each stage consumes the next unused terms of one sign, so every positive and every negative term eventually appears exactly once.

Let U_k be the partial sum at the end of the kth positive stage and D_k the partial sum at the end of the following negative stage. The last positive term is the only reason U_k crosses above 1, so

$$0 < U_k - 1 \leq \text{the last positive term used.}$$

Likewise,

$$0 < 1 - D_k \leq \text{the magnitude of the last negative term used.}$$

Because the terms of the alternating harmonic series tend to zero, both right-hand sides tend to zero. Hence

$$U_k \to 1, \qquad D_k \to 1.$$

Every intermediate partial sum lies between the two neighboring crossing values, so the entire rearranged partial-sum sequence converges to 1.

Thus this rearrangement has sum 1, whereas the original series has sum $L < 1$. The change is possible because the series is conditionally, not absolutely, convergent.

Connection. Absolute convergence makes the total tail mass small regardless of order. Conditional convergence controls only the signed cancellation, so reordering can change how that cancellation accumulates.

A tempting but false argument. Do not treat a convergence test as a name-recognition exercise. State its hypotheses, identify the model family, and explain the comparison being used.

Checkpoint. For each test in this chapter, identify whether its proof ultimately uses a geometric series, a p-series, or alternating cancellation.

Counterexample reflection.

Tempting claim, with context. For a real series $\sum a_i$, the condition $a_i \to 0$ is sufficient for convergence.

Pause. Small terms are necessary, but do they imply convergence of the partial sums?

Counterexample in this context. For $a_i = 1/i$, the terms tend to zero, yet the grouped harmonic partial sums grow without bound.

Exact failure and repaired claim. The Term Test is only necessary. A convergence proof needs additional structure such as comparison, geometric decay, or alternating cancellation under the stated hypotheses.

Counterexample reflection.

Tempting claim, with context. For a positive series, a Ratio- or Root-Test value equal to 1 determines convergence.

Pause. Compare two standard p-series whose ratio and root limits are both 1.

Counterexample in this context. Both $\sum 1/i$ and $\sum 1/i^2$ produce the value 1, but the first diverges and the second converges.

Exact failure and repaired claim. At the value 1, the ratio and root tests are inconclusive. Return to comparison with a model series.

Counterexample reflection.

Tempting claim, with context. For eventually nonnegative terms, if $0 \le a_i \le b_i$ and $\sum a_i$ converges, then $\sum b_i$ converges.

Pause. Comparison transfers information only in one direction. Which series is providing the upper bound?

Counterexample in this context. Take $a_i = 1/i^2$ and $b_i = 1/i$. Then $0 \le a_i \le b_i$, but $\sum a_i$ converges and $\sum b_i$ diverges.

Exact failure and repaired claim. A convergent majorant proves convergence of a smaller series; a divergent minorant proves divergence of a larger series. Reverse comparisons give no conclusion.

Return to the opening question. Grouping did not create convergence or divergence; it exposed how much mass each scale contributes. Absolute convergence is precisely the condition that makes regrouping and rearrangement harmless, while conditional convergence leaves order relevant.

Formative practice

Exercise 5.1. Fluency. Derive the finite geometric-sum formula and compute $\sum_{i=0}^{\infty} 3(1/4)^i$.

Exercise 5.2. Proof practice. Group the harmonic series into dyadic blocks and write the first four block lower bounds explicitly.

Exercise 5.3. Proof practice. Use the same grouping to prove $\sum 1/i^{3/2}$ converges.

Exercise 5.4. Proof practice. Use limit comparison with a p-series to classify $\sum (5i^2+1)/(i^4-3i)$.

Exercise 5.5. Proof practice. Use the Ratio Test to prove convergence of $\sum i/3^i$, and display the geometric tail estimate.

Exercise 5.6. Proof practice. Let $a_i = 2^{-i}\big(2 + \frac{1}{2}(-1)^i\big)$. Show that the ratio sequence $|a_{i+1}/a_i|$ has no limit, compute its lim sup, and apply the Ratio Test in its tail-envelope form.

Further problems

Exercise 5.7. Suppose $\sum x_i$ converges absolutely and (y_i) is bounded. Prove that $\sum x_i y_i$ converges absolutely. Then construct a counterexample when $\sum x_i$ is only conditionally convergent.

Exercise 5.8. Let $a_i > 0$. Prove that if $ia_i \to \ell \neq 0$, then $\sum a_i$ diverges, while if $i^2 a_i$ has a finite limit, then $\sum a_i$ converges. Identify the comparison series in each part.

Exercise 5.9. Use the Root Test to classify $\sum((2i+1)/(3i+2))^i$.

Exercise 5.10. Give one convergent and one divergent positive series for which the ratio and root tests return the value 1.

Exercise 5.11. Prove that $\sum(-1)^{i+1}/\sqrt{i}$ converges but not absolutely.

Exercise 5.12. For each series, justify the structural choice of test before carrying it out: $\sum 1/(i^2+7)$, $\sum i!/i^i$, and $\sum(-1)^i(i+1)/(i^2+1)$.

What should remain after this unit?
Write a brief answer in your own words.

One statement or distinction I can now formulate precisely:

One proof decision I can reuse:

One counterexample and the claim it repairs:

Open Sets, Closed Sets, and Limit Points

Suggested Abbott reading: Abbott §§3.1–3.2.
Chapter focus: Neighborhoods, open and closed sets, limit points, closure, and sequential characterizations.

Unit at a glance.

Main idea. Open sets encode local freedom under small perturbations; closed sets encode stability under limits.

Proof strategy. Negate "there exists a neighborhood" statements correctly.

Counterexample to remember. The principal counterexample and corrected claim are developed in the reflection before the exercises.

Prior knowledge. Intervals, sequences, and set operations.

Guiding question

How can we distinguish local stability under small perturbations from stability under limits?

Inquiry 6.1: Choose a topological representation *Detachable module*

Investigate. Classify

$$(0,1), \qquad [0,1], \qquad \mathbb{Z}, \qquad \{0\} \cup \{1/n : n \in \mathbb{N}\}, \qquad \mathbb{Q}$$

as open, closed, both, or neither in $\mathbb{R}$. For the last three sets, identify isolated points and limit points.

Decide and justify. For each set, choose a representation suited to the set: neighborhoods, complements, or convergent sequences. Do not use all three unless they genuinely add information.

Transfer. Construct a set that is neither open nor closed, has an isolated point, and has a limit point that is not contained in the set.

Stop here and record a first response. The resolution is absorbed into the core exposition that follows rather than repeated in a separate answer block.

Mathematics refresh. Interval endpoints encode membership: round parentheses omit an endpoint and square brackets include it. Complements turn endpoint information inside out; for example, $[0,1]^c = (-\infty, 0) \cup (1, \infty)$.

Review pointer: Review interval notation, complements, unions, intersections, and De Morgan's laws.

6.1 Neighborhoods and open sets

Two different stability questions. An open set gives each of its points some local room: sufficiently small perturbations remain inside the set. A closed set answers a different question: when points of the set converge, is the limiting point still admissible? For example, every point of $(0,1)$ has local room even though a sequence in $(0,1)$ may converge to an omitted endpoint, while $[0,1]$ contains such limiting endpoints even though its endpoints have no full neighborhood contained in the set.

Definition 6.1 (Neighborhood and open set)**.** For $x \in \mathbb{R}$ and $\varepsilon > 0$, the ε-neighborhood of x is

$$V_\varepsilon(x) = (x - \varepsilon, x + \varepsilon).$$

A set $O \subseteq \mathbb{R}$ is *open* if for every $x \in O$ there exists $\varepsilon > 0$ such that $V_\varepsilon(x) \subseteq O$.

Theorem 6.2. *Arbitrary unions of open sets are open, and finite intersections of open sets are open.*

Proof. If $x \in \bigcup_\alpha O_\alpha$, then $x \in O_{\alpha_0}$ for some index. Since O_{α_0} is open, one neighborhood of x lies in O_{α_0} and hence in the union. For a finite intersection, choose one neighborhood radius for each set and take the minimum. The finiteness is essential because the minimum of infinitely many positive radii may be zero. □

6.2 Limit points, isolated points, and closed sets

Definition 6.3 (Limit point)**.** A point x is a *limit point* of A if every deleted neighborhood

$$V_\varepsilon(x) \setminus \{x\}$$

meets A. A limit point need not belong to A.

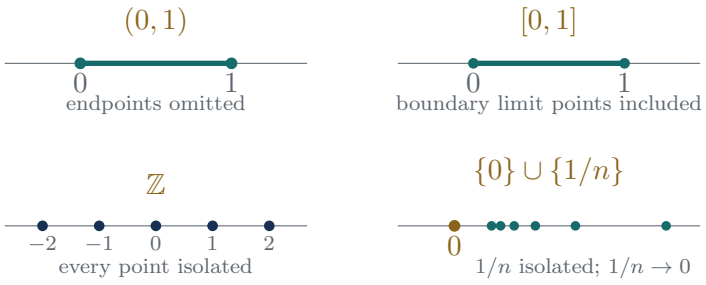
Definition 6.4 (Isolated point)**.** A point $x \in A$ is *isolated in* A if some neighborhood contains no other point of A:

$$\exists \varepsilon > 0, \qquad V_\varepsilon(x) \cap A = \{x\}\,.$$

The notions are complementary for points of the set: every $x \in A$ is either isolated in A or is a limit point of A, but not both. A point outside A can be a limit point, but it cannot be an isolated point of A. For

$$A = \{1/n : n \in \mathbb{N}\}\,,$$

every $1/n$ is isolated, while 0 is a limit point not contained in A.



Four recurring topological patterns: omitted boundary points, included boundary points, isolated points, and isolated points accumulating at an included limit point.

Definition 6.5 (Closed set)**.** A set $F \subseteq \mathbb{R}$ is *closed* if it contains all of its limit points.

Theorem 6.6 (Open–closed duality)**.** *For a set $F \subseteq \mathbb{R}$, the following are equivalent:*

(i) F is closed;

(ii) F^c is open.

Consequently, arbitrary intersections of closed sets are closed, and finite unions of closed sets are closed.

Pause and reconstruct: Open–closed duality

Before continuing. Prove that $F \subseteq \mathbb{R}$ is closed if and only if F^c is open.

Guided step. For the forward direction, take $x \in F^c$ and suppose no neighborhood of x lies in F^c; explain why this makes x a limit point of F. For the reverse direction, let x be a limit point of F and suppose $x \notin F$; use openness of F^c to obtain the contradiction.

Check. Which definition of closedness is being used? How would the argument change if closedness were defined sequentially?

Proof. Assume first that F is closed, and let $x \in F^c$. If every neighborhood of x met F, then, because $x \notin F$, every deleted neighborhood of x would meet F. Thus x would be a limit point of F, contradicting closedness. Hence some neighborhood of x is disjoint from F, and therefore lies in F^c. So F^c is open.

Conversely, assume F^c is open, and let x be a limit point of F. If $x \notin F$, then $x \in F^c$, so some neighborhood of x lies in F^c and is disjoint from F. This contradicts the definition of a limit point. Hence $x \in F$, and F is closed.

The final statements follow from De Morgan's laws and Theorem 6.2. □

Theorem 6.7 (Sequential characterization)**.** *A set $F \subseteq \mathbb{R}$ is closed if and only if every convergent sequence $x_n \in F$ has its limit in F.*

Proof. Assume F is closed and $x_n \in F$ with $x_n \to x$. If $x \notin F$, then the open set F^c contains a neighborhood of x, which eventually contains x_n, a contradiction.

Conversely, suppose every convergent sequence in F has its limit in F. If x is a limit point of F, choose $x_n \in F \cap (V_{1/n}(x) \setminus \{x\})$. Then $x_n \to x$, so $x \in F$. □

6.3 Closure, interior, and boundary

Definition 6.8 (Closure, interior, and boundary)**.** The *closure* $\mathrm{cl}(A)$ is A together with all its limit points. The *interior* $\mathrm{int}(A)$ is the set of points possessing a neighborhood contained in A. The *boundary* is

$$\partial A = \mathrm{cl}(A) \setminus \mathrm{int}(A).$$

For $A = (0,1) \cap \mathbb{Q}$,

$$\mathrm{int}(A) = \varnothing, \qquad \mathrm{cl}(A) = [0,1], \qquad \partial A = [0,1].$$

Density makes every point of $[0,1]$ a limit point, but no interval lies inside A.

Worked example 6.9 (The Cantor set, first view)**.** Start with $C_0 = [0,1]$. Remove the open middle third to obtain C_1. Repeat on every remaining interval. The Cantor set is

$$C = \bigcap_{n=0}^{\infty} C_n.$$

Each C_n is closed, so C is closed. Unit 7 will prove that C contains no nontrivial interval and will connect this with the uncountability phenomenon established in Theorem 2.17.

A tempting but false argument. Saying a set is closed because it contains its endpoints. That phrase only applies to intervals; closedness means containing every limit point.

Checkpoint. Classify each set as open, closed, both, or neither: $\mathbb{R}$, $\varnothing$, $[0,1)$, $\mathbb{Z}$, and $\mathbb{Q}$.

Counterexample reflection.

Tempting claim, with context. In $\mathbb{R}$, an arbitrary intersection of open sets is open.

Pause. Finite intersections preserve a positive common neighborhood radius. What if the available radii shrink to zero?

Counterexample in this context. $\bigcap_{n=1}^{\infty}(-1/n, 1/n) = \{0\}$, and $\{0\}$ is not open in $\mathbb{R}$.

Exact failure and repaired claim. Only finite intersections of open sets are guaranteed to be open; arbitrary unions are open.

Counterexample reflection.

Tempting claim, with context. Every limit point of a set $A \subseteq \mathbb{R}$ must belong to A.

Pause. Compare the definition of limit point with the definition of closedness.

Counterexample in this context. For $A = \{1/n : n \in \mathbb{N}\}$, the point $0 \notin A$ is a limit point.

Exact failure and repaired claim. Membership is not part of the limit-point definition. A set contains all of its limit points exactly when it is closed.

Counterexample reflection.

Tempting claim, with context. Every point of an infinite subset of $\mathbb{R}$ is a limit point of that set.

Pause. Infinitude is global. Does it prevent an individual point from having a neighborhood containing no other set points?

Counterexample in this context. The set $\mathbb{Z}$ is infinite, but every integer is isolated: for $m \in \mathbb{Z}$, the neighborhood $V_{1/2}(m)$ meets $\mathbb{Z}$ only at m.

Exact failure and repaired claim. An infinite set may contain isolated points. Limit-point behavior requires arbitrarily nearby distinct points, not merely infinitely many points somewhere in the set.

Formative practice

Exercise 6.1. Fluency. For $A = \{1/n : n \in \mathbb{N}\}$, identify every isolated point and every limit point. Explain why 0 belongs to one list but not the other.

Exercise 6.2. Proof practice. Prove directly that (a, b) is open and $[a, b]$ is closed.

Exercise 6.3. Counterexample reflection. Give an infinite family of closed sets whose union is not closed.

Exercise 6.4. Proof practice. Let A' denote the set of limit points of A. Using the definition $\mathrm{cl}(A) = A \cup A'$, prove that $\mathrm{cl}(A)$ is closed and that every closed set containing A also contains $\mathrm{cl}(A)$. Conclude that $\mathrm{cl}(A)$ is the smallest closed set containing A.

Further problems

Exercise 6.5. Show that $x \in \mathrm{cl}(A)$ if and only if there exists a sequence $a_n \in A$ with $a_n \to x$.

Exercise 6.6. Prove that the boundary $\partial A = \mathrm{cl}(A) \setminus \mathrm{int}(A)$ is closed. Choose one of complements, sequences, or set identities, and justify why it gives a direct proof.

Exercise 6.7. Prove that a set is both open and closed in $\mathbb{R}$ only if it is $\varnothing$ or $\mathbb{R}$. You may defer the connectedness step until Unit 7, but identify exactly what remains to be proved.

Exercise 6.8 (Enrichment)**.** Let $O \subseteq \mathbb{R}$ be nonempty and open. For each $x \in O$, let I_x be the union of all open intervals containing x and contained in O. Prove that each I_x is an open interval, that two distinct maximal intervals are disjoint, and that there are only countably many of them by assigning a rational number to each. Conclude that every open subset of $\mathbb{R}$ is a countable union of disjoint open intervals.

What should remain after this unit?
Write a brief answer in your own words.

One statement or distinction I can now formulate precisely:

One proof decision I can reuse:

One counterexample and the claim it repairs:

Additional Problems

The following probems are grouped by topic, and no particular problem is required for the continuity of the exposition in the main 14 units.

A.1 Logic, completeness, and countability

Exercise A.1 (Irrationality extensions)**.** (a) Prove that $\sqrt{3}$ is irrational.

(b) Adapt the argument to prove that $\sqrt{6}$ is irrational.

(c) Explain exactly where the same proof strategy fails if one attempts to prove that $\sqrt{4}$ is irrational.

Exercise A.2 (Finite and countable De Morgan laws)**.** Let (A_n) be a sequence of subsets of $\mathbb{R}$.

(a) Use induction to prove, for every $N \in \mathbb{N}$,

$$\left(\bigcup_{n=1}^{N} A_n\right)^c = \bigcap_{n=1}^{N} A_n^c.$$

(b) Explain why finite induction alone cannot establish the corresponding identity for a countably infinite union.

(c) Prove directly, by testing membership of an arbitrary point, that

$$\left(\bigcup_{n=1}^{\infty} A_n\right)^c = \bigcap_{n=1}^{\infty} A_n^c.$$

State and prove the companion identity for countable intersections.

Exercise A.3 (Two consequences of the supremum definition)**.** Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above.

(a) Suppose a is an upper bound for A and $a \in A$. Prove that $a = \max A = \sup A$.

(b) Suppose $\sup A < \sup B$. Prove that there exists $b \in B$ that is an upper bound for A.

Exercise A.4 (Enumerating an infinite subset)**.** Let B be countable, let $f : \mathbb{N} \to B$ be a bijection, and let $A \subseteq B$ be infinite. Define

$$n_1 = \min\{n \in \mathbb{N} : f(n) \in A\}.$$

Inductively define n_{k+1} as the least integer $n > n_k$ for which $f(n) \in A$, and set $g(k) = f(n_k)$. Prove that every n_k is well defined, and that $g : \mathbb{N} \to A$ is a bijection. Conclude that every infinite subset of a countable set is countable.

A.2 Sequences and series

Exercise A.5 (Counterexamples for algebraic limit claims)**.** For each request, either construct sequences with the stated properties or prove that no such example exists.

(a) Both (x_n) and (y_n) diverge, but $(x_n + y_n)$ converges.

(b) (x_n) converges, (y_n) diverges, and $(x_n + y_n)$ converges.

(c) (b_n) converges and $b_n \neq 0$ for every n, but $(1/b_n)$ diverges.

(d) (a_n) is unbounded, (b_n) converges, and $(a_n - b_n)$ is bounded.

(e) Both $(a_n b_n)$ and (a_n) converge, but (b_n) diverges.

For each impossible request, identify the applicable limit law.

Exercise A.6 (Subsequence construction bank)**.** For each request, either construct a sequence or prove that the request is impossible.

(a) A sequence containing neither 0 nor 1 as a term, but having one subsequence converging to 0 and another converging to 1.

(b) A monotone divergent sequence with a convergent subsequence.

(c) A sequence having a subsequence converging to each point of $\{1, 1/2, 1/3, \ldots\}$.

(d) An unbounded sequence with a convergent subsequence.

(e) A sequence having a bounded subsequence that contains no convergent subsequence.

Exercise A.7 (Cauchy counterexample bank)**.** For each request, either construct a sequence or prove that the request is impossible.

(a) A Cauchy sequence that is not monotone.

(b) A monotone sequence that is not Cauchy.

(c) A Cauchy sequence with a divergent subsequence.

(d) An unbounded sequence containing a Cauchy subsequence.

Exercise A.8 (Transforming convergent series)**.** (a) Prove that if $\sum a_n$ converges absolutely, then $\sum a_n^2$ converges absolutely. Determine whether absolute convergence can be replaced by ordinary convergence.

(b) Suppose $a_n \geq 0$ and $\sum a_n$ converges. Determine what, if anything, must follow about $\sum \sqrt{a_n}$. Support the conclusion with proofs or counterexamples.

A.3 Topology, compactness, and continuity

Exercise A.9 (Open and closed set diagnostics)**.** Decide whether each statement is true. Prove every true statement and give a counterexample to every false statement.

(a) For every set $A \subseteq \mathbb{R}$, the complement A^c is open.

(b) A set containing an isolated point cannot be open.

(c) A set A is closed if and only if $A = \overline{A}$.

(d) If A is closed and B is open, then $A \setminus B$ is closed.

(e) If A is bounded above, then $\sup A$ is a limit point of A.

(f) Every finite subset of $\mathbb{R}$ is closed.

(g) An open set containing every rational number must equal $\mathbb{R}$.

Exercise A.10 (Compactness diagnostics)**.** Decide whether each statement is true. Prove every true statement and give a counterexample to every false statement.

(a) The intersection of any nonempty family of compact subsets of $\mathbb{R}$ is compact.

(b) If $A \subseteq \mathbb{R}$ is arbitrary and $K \subseteq \mathbb{R}$ is compact, then $A \cap K$ is compact.

(c) If $F_1 \supseteq F_2 \supseteq \cdots$ is a nested sequence of nonempty closed subsets of $\mathbb{R}$, then $\bigcap_{n=1}^{\infty} F_n$ is nonempty.

(d) Every finite subset of $\mathbb{R}$ is compact.

(e) Every countable subset of $\mathbb{R}$ is compact.

When a statement fails, identify the missing hypothesis that would repair it.

Exercise A.11 (Continuity of the cube-root function)**.** Let $g(x) = \sqrt[3]{x}$.

(a) Prove directly that g is continuous at 0.

(b) Let $c \neq 0$. Use
$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$
to prove directly that g is continuous at c.

Make every local lower bound in the denominator explicit.

A.4 Mean value methods and smoothness

Exercise A.12 (Strict convexity and an interior fixed point)**.** Let g be twice differentiable on an open interval containing $[0,1]$, with $g''(x) > 0$ for $0 < x < 1$, $g(0) > 0$, and $g(1) = 1$. Prove the equivalence
$$\text{there exists } d \in (0,1) \text{ with } g(d) = d \quad \Longleftrightarrow \quad g'(1) > 1.$$

A complete proof should identify where strict convexity, the Intermediate Value Theorem, and Rolle's theorem enter.

Exercise A.13 (Hallmark extension: smooth but nonanalytic)**.** Define
$$f(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

You may use the fact that $t^m e^{-t} \to 0$ as $t \to \infty$ for every nonnegative integer m.

(a) Compute $f'(x)$ for $x \neq 0$ and prove from the definition that $f'(0) = 0$.

(b) Compute $f''(x)$ for $x \neq 0$ and prove that $f''(0) = 0$.

(c) Prove by induction that, for every $r \in \mathbb{N}$, there is a polynomial P_{2r} of degree at most $2r$ such that
$$f^{(r)}(x) = \begin{cases} P_{2r}(1/x)e^{-1/x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$
Conclude that f is infinitely differentiable on $\mathbb{R}$.

(d) Determine the Maclaurin series of f and prove that it does not agree with f on any neighborhood of 0.

A.5 Integral properties and the Fundamental Theorem

Exercise A.14 (Positivity and the integral)**.** Decide whether each statement is true. Prove every true statement and give a counterexample to every false statement.

(a) If $|f|$ is Riemann integrable on $[a, b]$, then f is Riemann integrable there.

(b) If g is Riemann integrable, $g \geq 0$, and $g(x) > 0$ at infinitely many points, then $\int_a^b g > 0$.

(c) If g is continuous, $g \geq 0$, and $g(x_0) > 0$ for at least one point x_0, then $\int_a^b g > 0$.

(d) If $\int_a^b f > 0$, then there are an interval $[c, d] \subseteq [a, b]$ and a number $\delta > 0$ such that $f(x) \geq \delta$ for every $x \in [c, d]$.

Exercise A.15 (Average value theorem for integrals)**.** Let g be continuous on $[a, b]$. Prove that there exists $c \in (a, b)$ such that

$$g(c) = \frac{1}{b-a} \int_a^b g(x)\, dx.$$

Give one proof using the accumulation form of the Fundamental Theorem and the Mean Value Theorem, and explain how the Extreme Value Theorem gives an alternative proof.

Exercise A.16 (Integration by parts and its hypotheses)**.** (a) Suppose F and G are differentiable on an open interval containing $[a, b]$, and suppose F' and G' are continuous on $[a, b]$. Prove

$$\int_a^b F(x)G'(x)\, dx = F(b)G(b) - F(a)G(a) - \int_a^b F'(x)G(x)\, dx.$$

Identify the product rule, product integrability, and the Fundamental Theorem at their exact points of use.

(b) Define

$$F(x) = \mathbf{1}_{[1,4]}(x), \qquad G(x) = \mathbf{1}_{[2,5]}(x).$$

At every point where the derivatives exist, one has $F' = G' = 0$. A formal substitution of these partial derivative data into the integration-by-parts identity from part (a), now on $[0, 3]$, gives a contradiction. Explain precisely why the hypotheses of part (a) are not satisfied and why derivative identities that fail at the jump points cannot replace them in this Riemann-integral theorem.

Proof-Building Toolkit

B.1 Logic and quantifier reference

Statement	Negation
$P \wedge Q$	$(\neg P) \vee (\neg Q)$
$P \vee Q$	$(\neg P) \wedge (\neg Q)$
$P \Rightarrow Q$	$P \wedge \neg Q$
$\forall x \in D:\ P(x)$	$\exists x \in D:\ \neg P(x)$
$\exists x \in D:\ P(x)$	$\forall x \in D:\ \neg P(x)$

B.2 Quantifier templates

Concept	Quantifier form
Sequence convergence	$\forall \varepsilon > 0\ \exists N\ \forall n \geq N:\ \lvert a_n - L\rvert < \varepsilon$
Sequence nonconvergence to L	$\exists \varepsilon_0 > 0\ \forall N\ \exists n \geq N:\ \lvert a_n - L\rvert \geq \varepsilon_0$
Functional limit	$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x:\ 0 < \lvert x - c\rvert < \delta \Rightarrow \lvert f(x) - L\rvert < \varepsilon$
Continuity at c	$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x:\ \lvert x - c\rvert < \delta \Rightarrow \lvert f(x) - f(c)\rvert < \varepsilon$
Uniform continuity	$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y:\ \lvert x - y\rvert < \delta \Rightarrow \lvert f(x) - f(y)\rvert < \varepsilon$
Cauchy sequence	$\forall \varepsilon > 0\ \exists N\ \forall m > n \geq N:\ \lvert a_m - a_n\rvert < \varepsilon$
Open set	$\forall x \in O\ \exists \varepsilon > 0:\ V_\varepsilon(x) \subseteq O$
Limit point	$\forall \varepsilon > 0:\ (V_\varepsilon(x) \setminus \{x\}) \cap A \neq \varnothing$

B.3 Standard proof openings

- Universal claim: "Let x be arbitrary in the stated domain."
- Existence claim: "We construct x as follows," followed by verification.
- Equality of sets: prove both inclusions.
- Limit claim: "Let $\varepsilon > 0$," then construct the threshold.
- Closed-set claim: use complements or a convergent sequence contained in the set.
- Compactness claim in $\mathbb{R}$: check closedness and boundedness; use sequential compactness first, and open covers when a finite subcover is needed.
- Non-uniformity: find x_n, y_n with $\lvert x_n - y_n\rvert \to 0$ but outputs stay separated.

B.4 Hypothesis ledger

Before invoking a theorem, write its hypotheses in a short ledger. For the Mean Value Theorem:

required	where checked
f continuous on $[a, b]$	line or prior theorem
f differentiable on (a, b)	line or prior theorem
$a < b$	declared interval

This habit prevents silent use of compactness, interval structure, boundedness, or denominator nonvanishing.

Counterexample Atlas

False claim	Counterexample	Missing idea
Bounded sequence $\Rightarrow$ convergent	$(-1)^n$	Need monotonicity or a subsequence conclusion
$a_n \to 0 \Rightarrow \sum a_n$ converges	$a_n = 1/n$	Terms vanishing is only necessary
Infinite intersection of open sets is open	$\bigcap_n (-1/n, 1/n) = \{0\}$	Only finite intersections are guaranteed
Closed $\Rightarrow$ compact in $\mathbb{R}$	$\mathbb{R}$	Need boundedness
Bounded $\Rightarrow$ compact	$(0, 1)$	Need closedness
Continuous $\Rightarrow$ uniformly continuous	x^2 on $\mathbb{R}$	Compactness or another global bound
Continuous $\Rightarrow$ differentiable	$\lvert x\rvert$ at 0	Difference quotients may disagree
Differentiable $\Rightarrow$ f' continuous	$x^2 \sin(1/x)$ with $f(0) = 0$	Derivatives have Darboux, not necessarily continuity
$f' = 0 \Rightarrow f$ constant on any domain	Piecewise constants on disconnected components	Need interval/connected domain
Bounded $\Rightarrow$ Riemann integrable	Dirichlet function	Oscillation remains maximal on every interval
Integrable $\Rightarrow$ continuous	Step function	Discontinuities may occupy a negligible set
IVP $\Rightarrow$ continuous	$g(0) = 0$, $g(x) = \sin(1/x)$ for $x \neq 0$	IVP rules out jumps, not oscillatory discontinuity
IVP $\Rightarrow$ monotone	$\sin x$	Attaining intermediate values does not impose an order direction

End-of-Course Proof Checklist

Before submitting a proof, verify:

1. Have all arbitrary objects and domains been declared?
2. Are the quantifiers in the correct order?
3. Does each chosen N, δ, subsequence, cover element, or partition depend only on previously fixed data?
4. Were all theorem hypotheses checked where the theorem is invoked?
5. If a supremum is used, did the proof distinguish "upper bound" from "attained maximum"?
6. If compactness is used, is the limit shown to remain in the set?
7. If a statement is false, does the counterexample satisfy every retained hypothesis and violate exactly the conclusion?
8. Does the final sentence explicitly return to the claim being proved?

References and Source Roles

[1] Stephen Abbott, *Understanding Analysis*, 2nd ed., Springer, 2015. The primary structural reference for Chapters 1–5 and 7, including its motivating investigations and theorem sequence.

The present manuscript is an original course synthesis. It follows Abbott's topic sequence selectively but does not reproduce the textbook's prose or exercise collection. Its distinctive features are explicit proof strategies, hypothesis ledgers, embedded proof reconstruction, and a systematic counterexample atlas.